

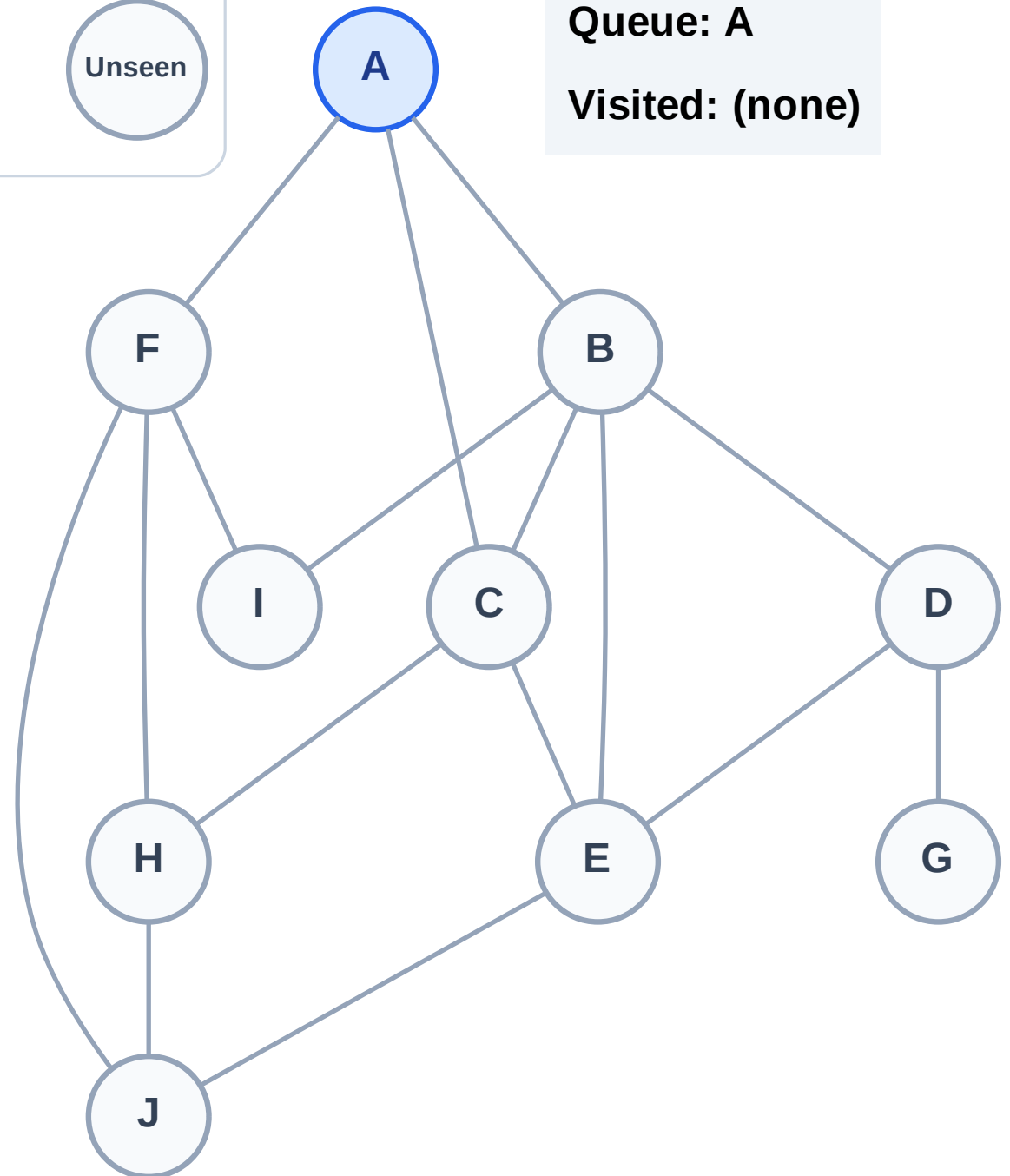
BFS Traversal

Step 1: Start at A

Legend



Queue: A
Visited: (none)



BFS Traversal

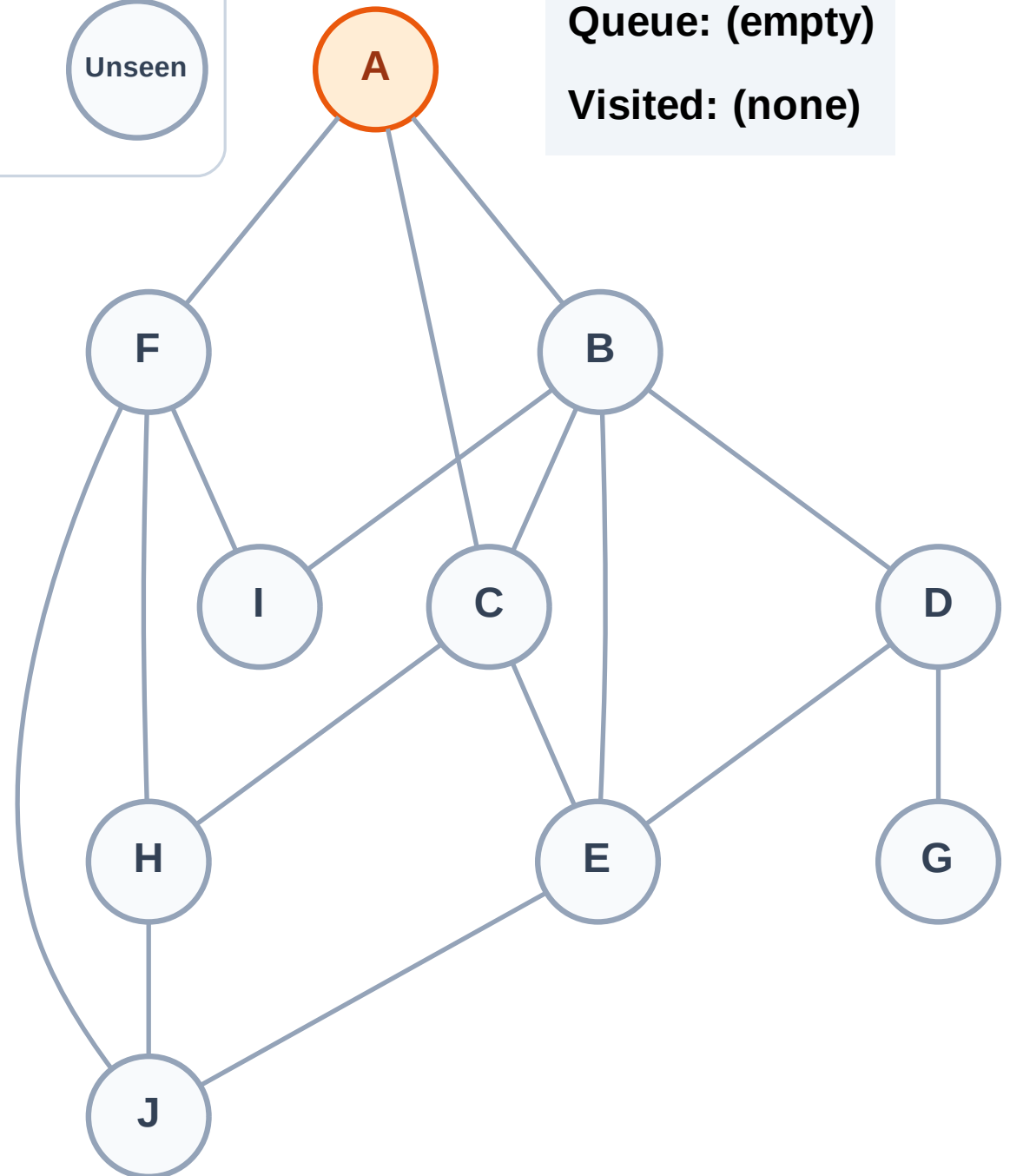
Step 2: Process node A

Legend



Queue: (empty)

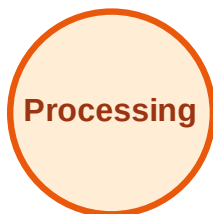
Visited: (none)



BFS Traversal

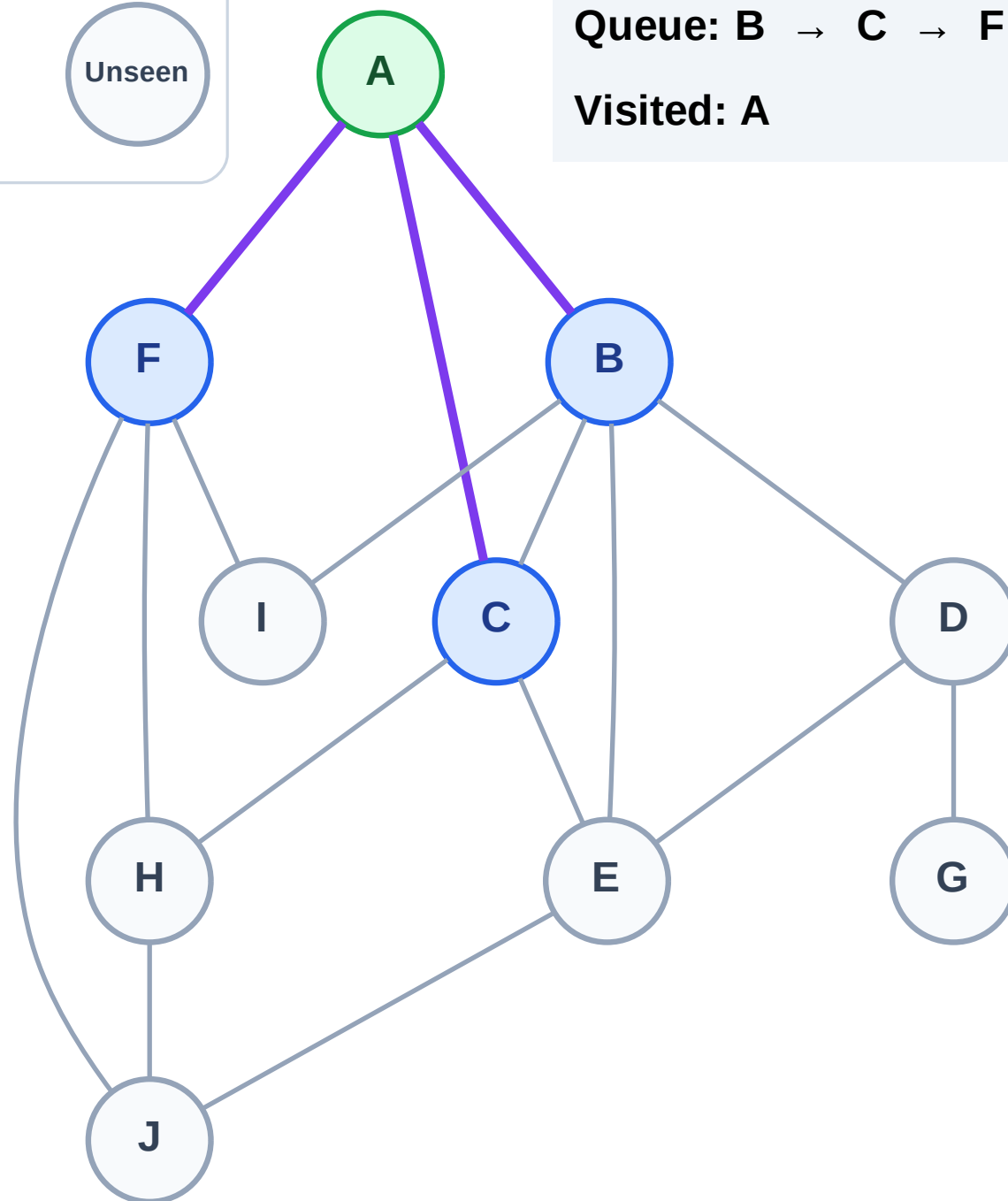
Step 3: Discover B, C, F from A

Legend



Queue: B → C → F

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

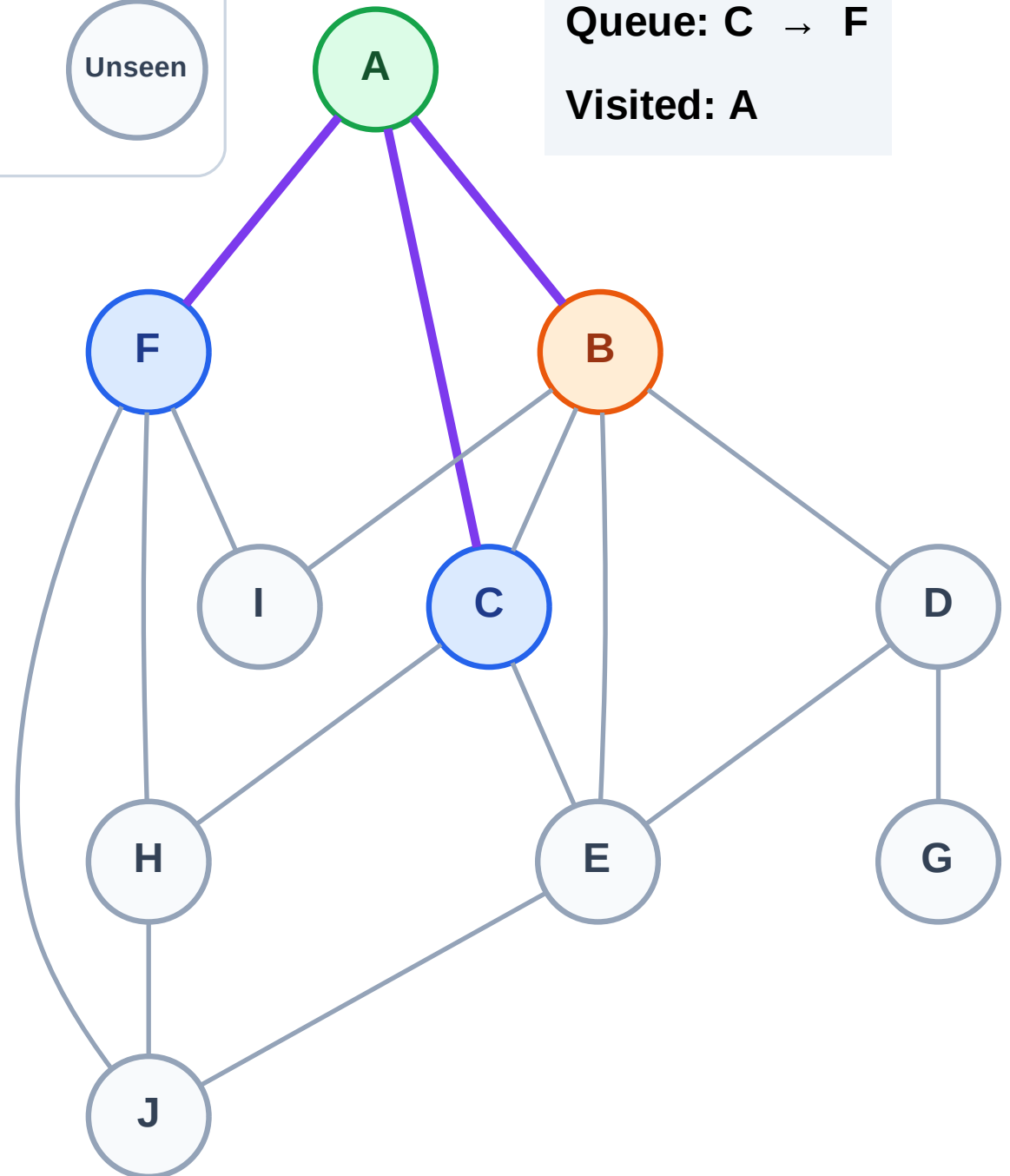
Processing

Discovered

Unseen

Queue: C → F

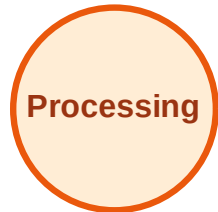
Visited: A



BFS Traversal

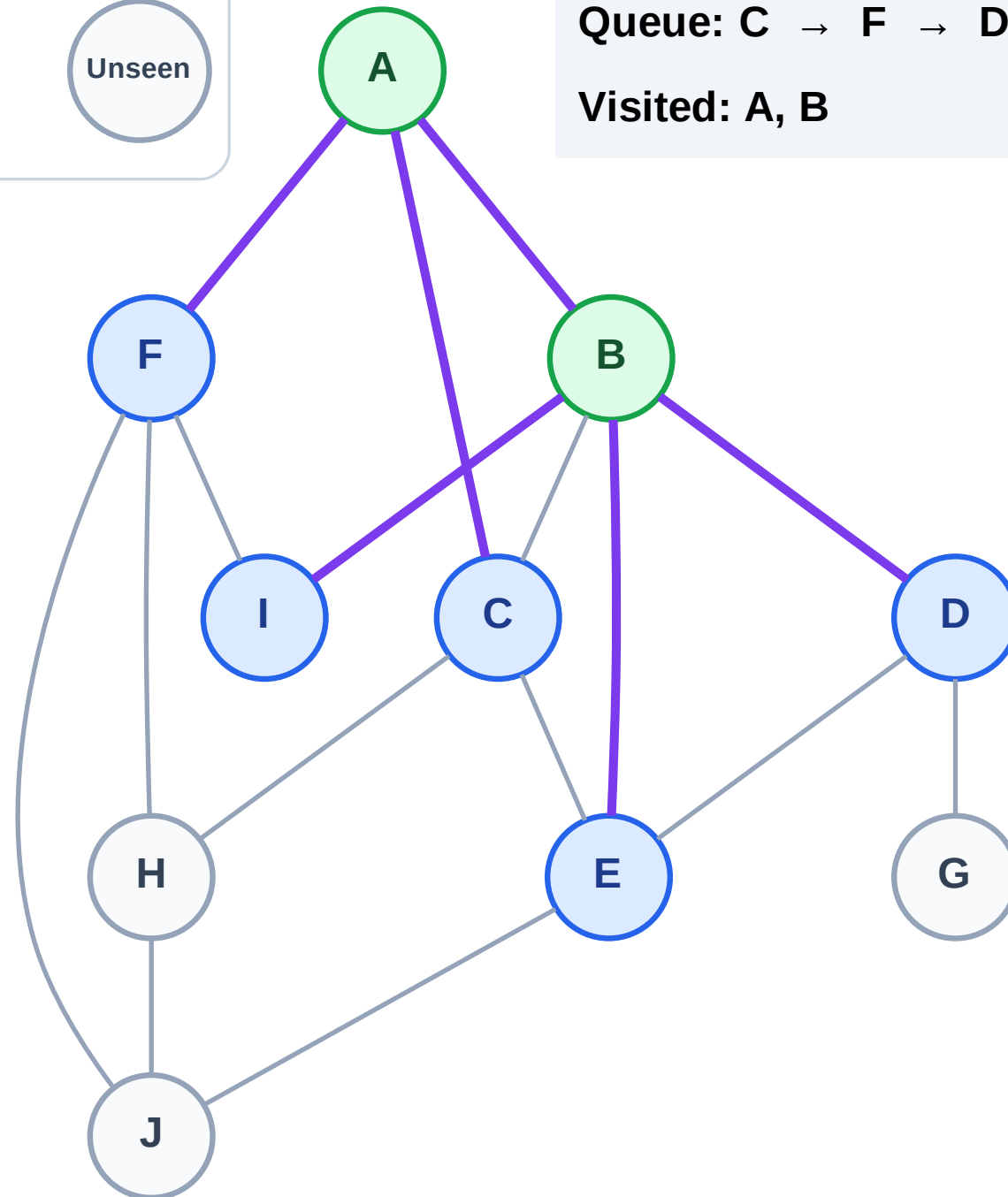
Step 5: Discover D, E, I from B

Legend



Queue: C → F → D → E → I

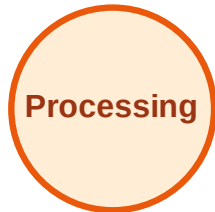
Visited: A, B



BFS Traversal

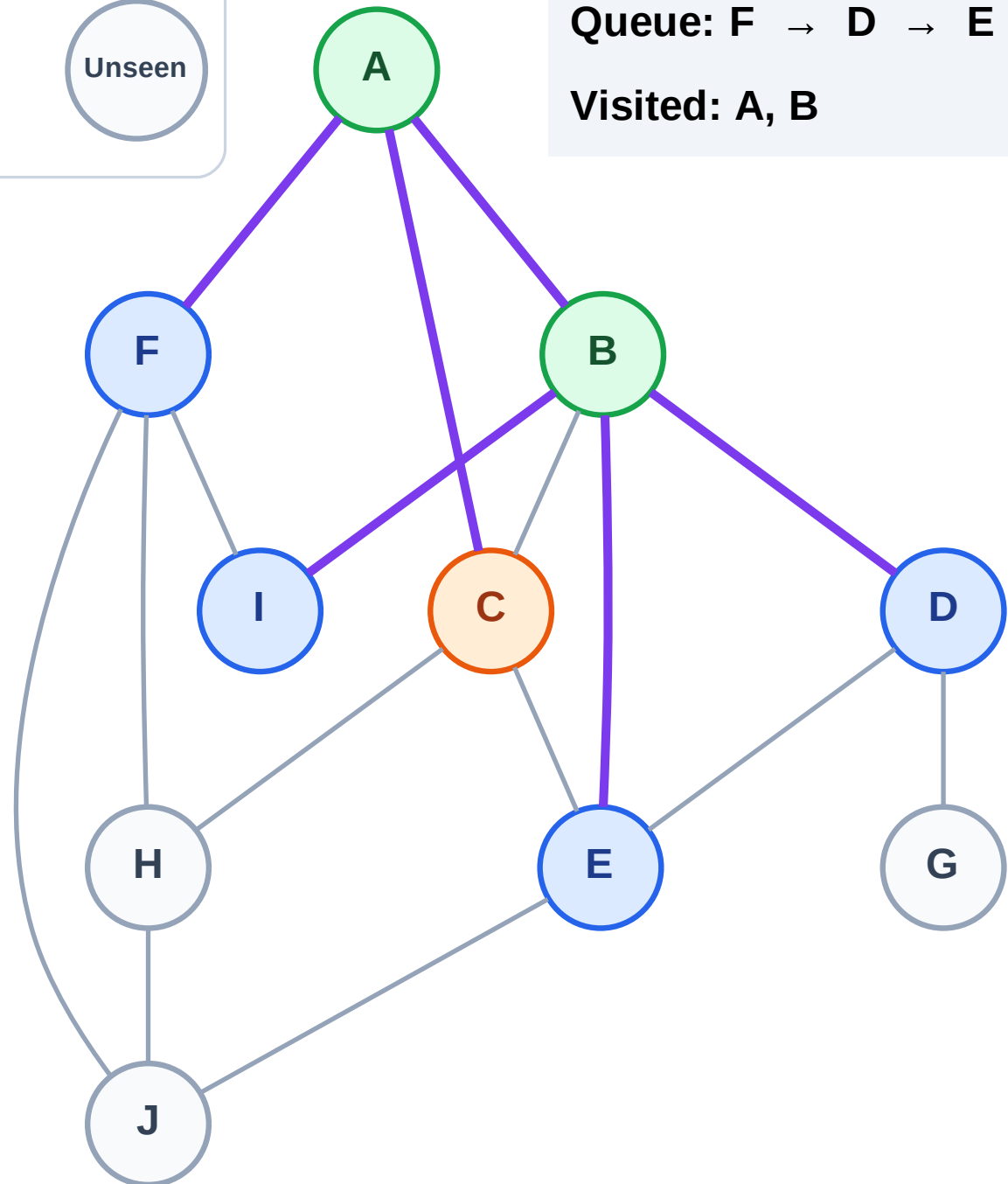
Step 6: Process node C

Legend



Queue: F → D → E → I

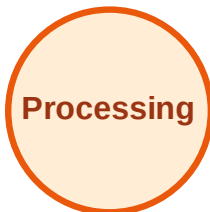
Visited: A, B



BFS Traversal

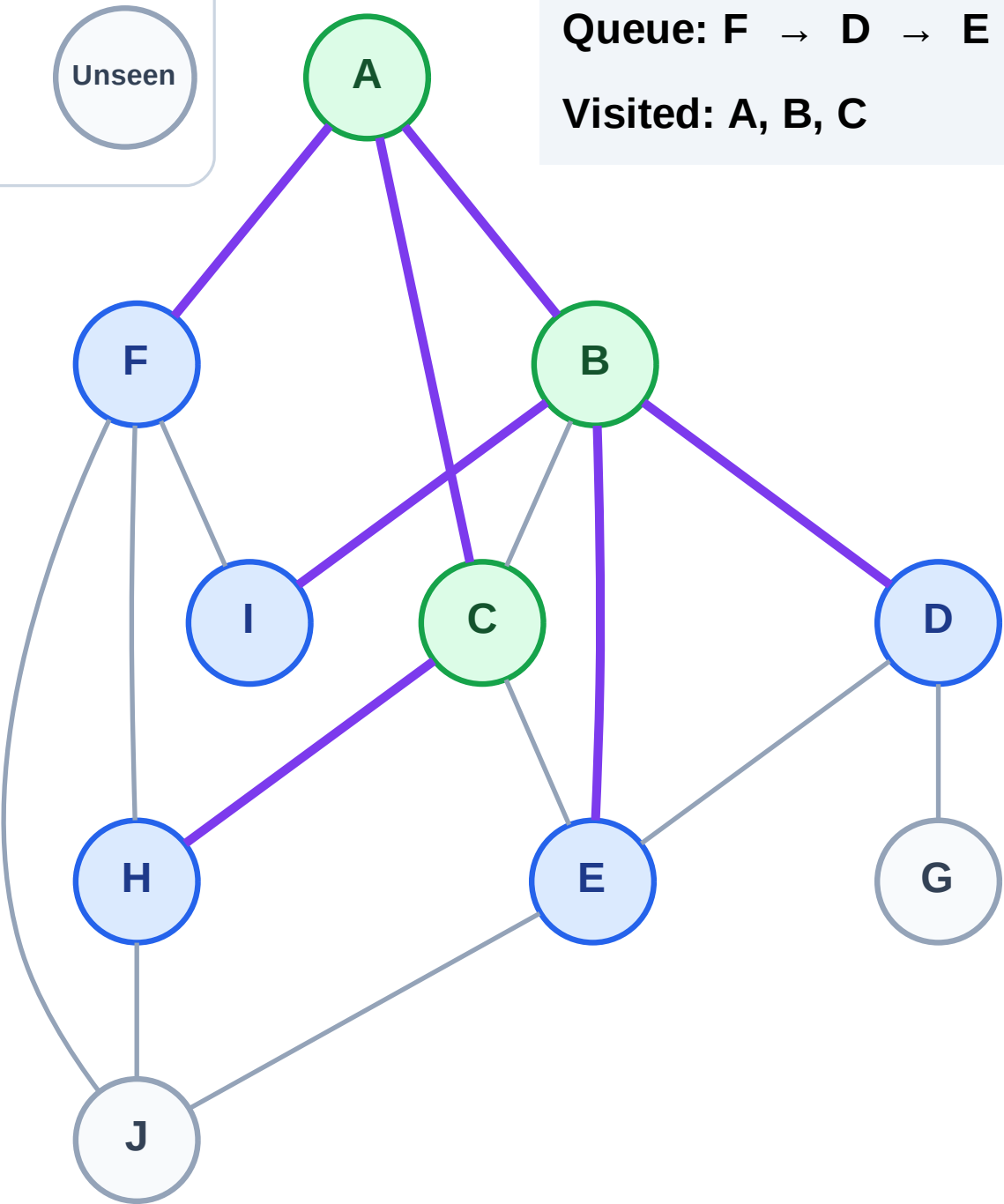
Step 7: Discover H from C

Legend



Queue: F → D → E → I → H

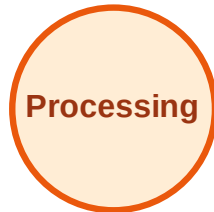
Visited: A, B, C



BFS Traversal

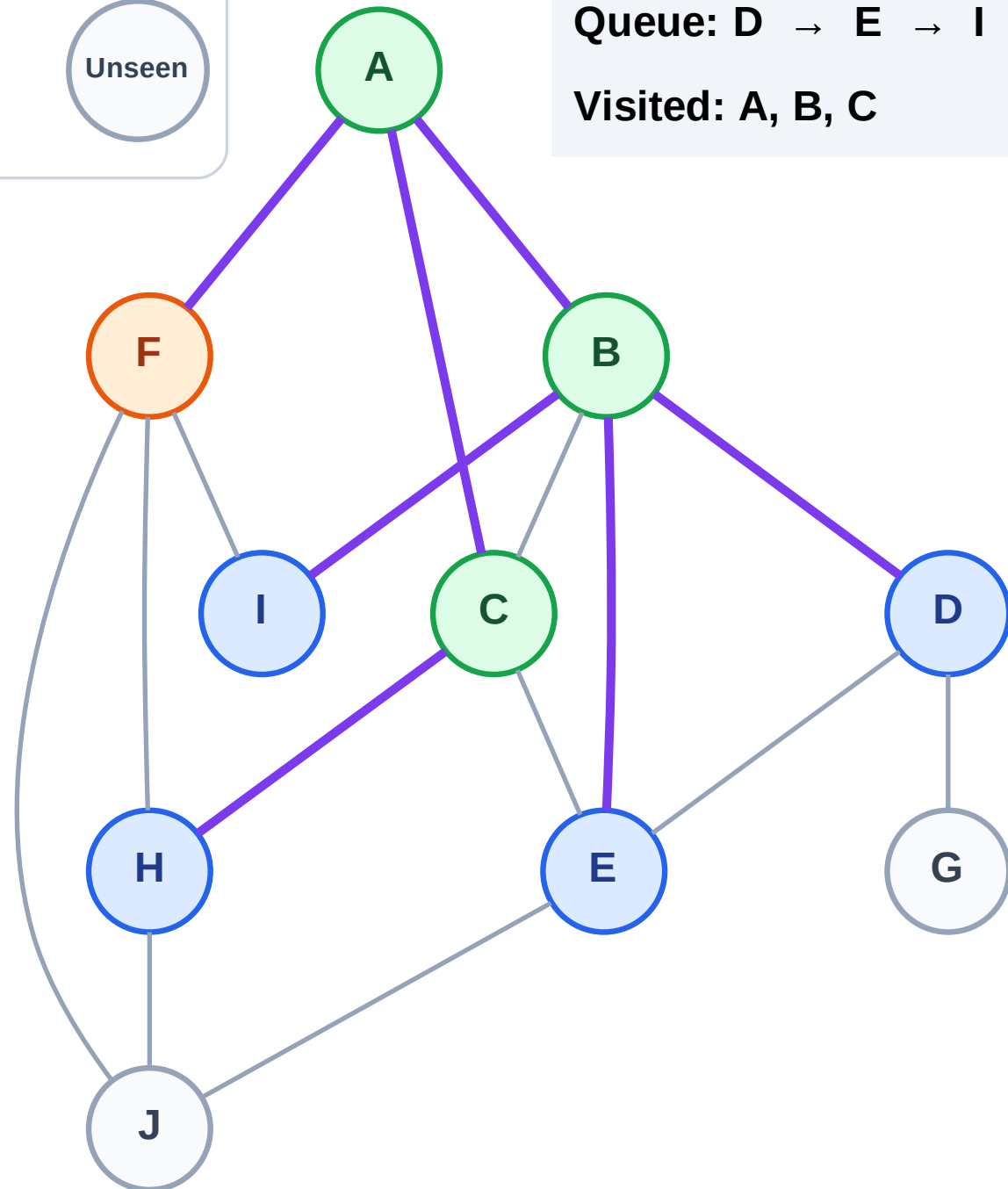
Step 8: Process node F

Legend



Queue: D → E → I → H

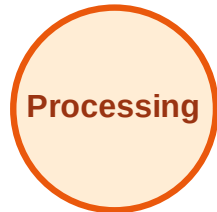
Visited: A, B, C



BFS Traversal

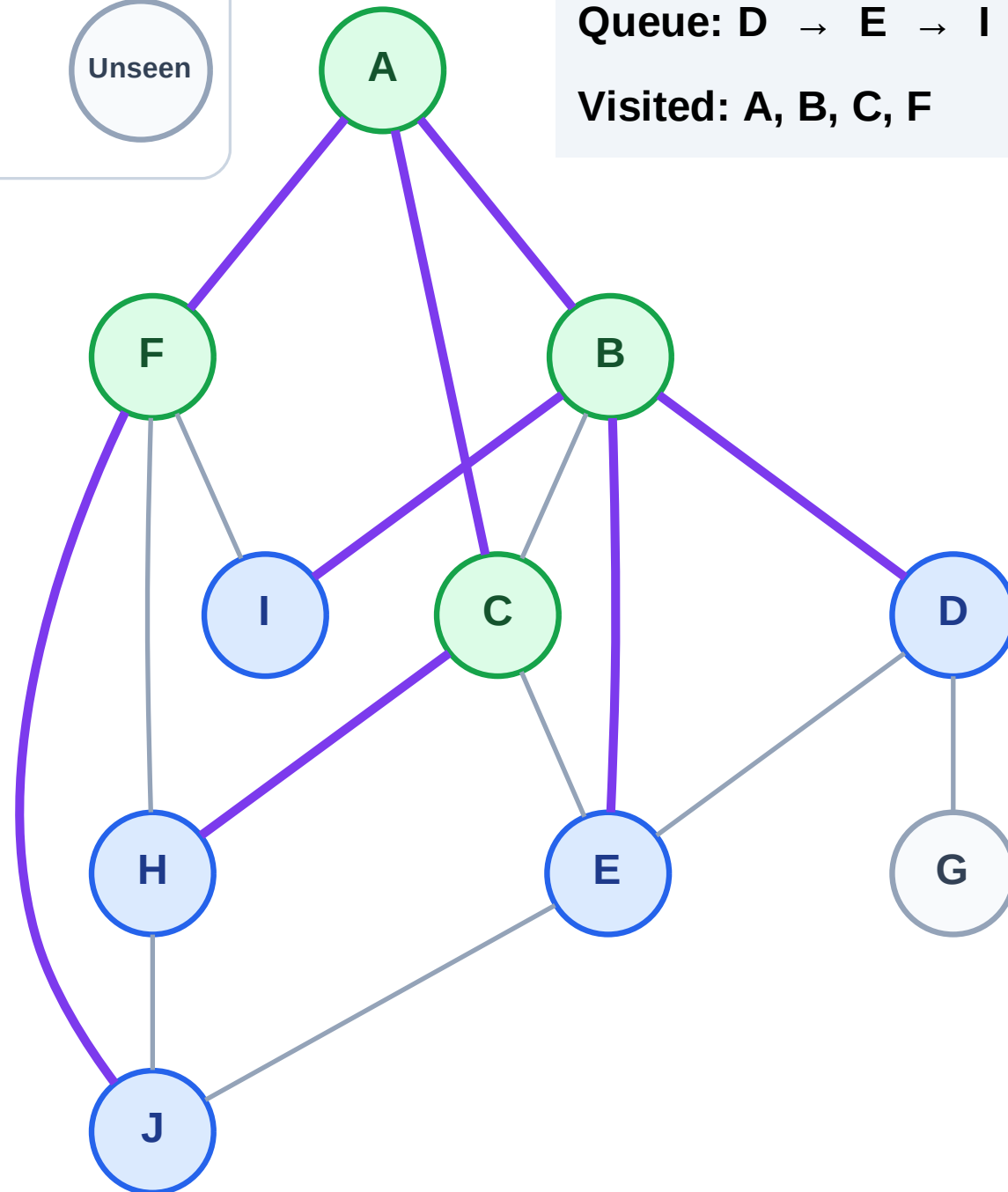
Step 9: Discover J from F

Legend



Queue: D → E → I → H → J

Visited: A, B, C, F



BFS Traversal

Step 10: Process node D

Legend

Complete

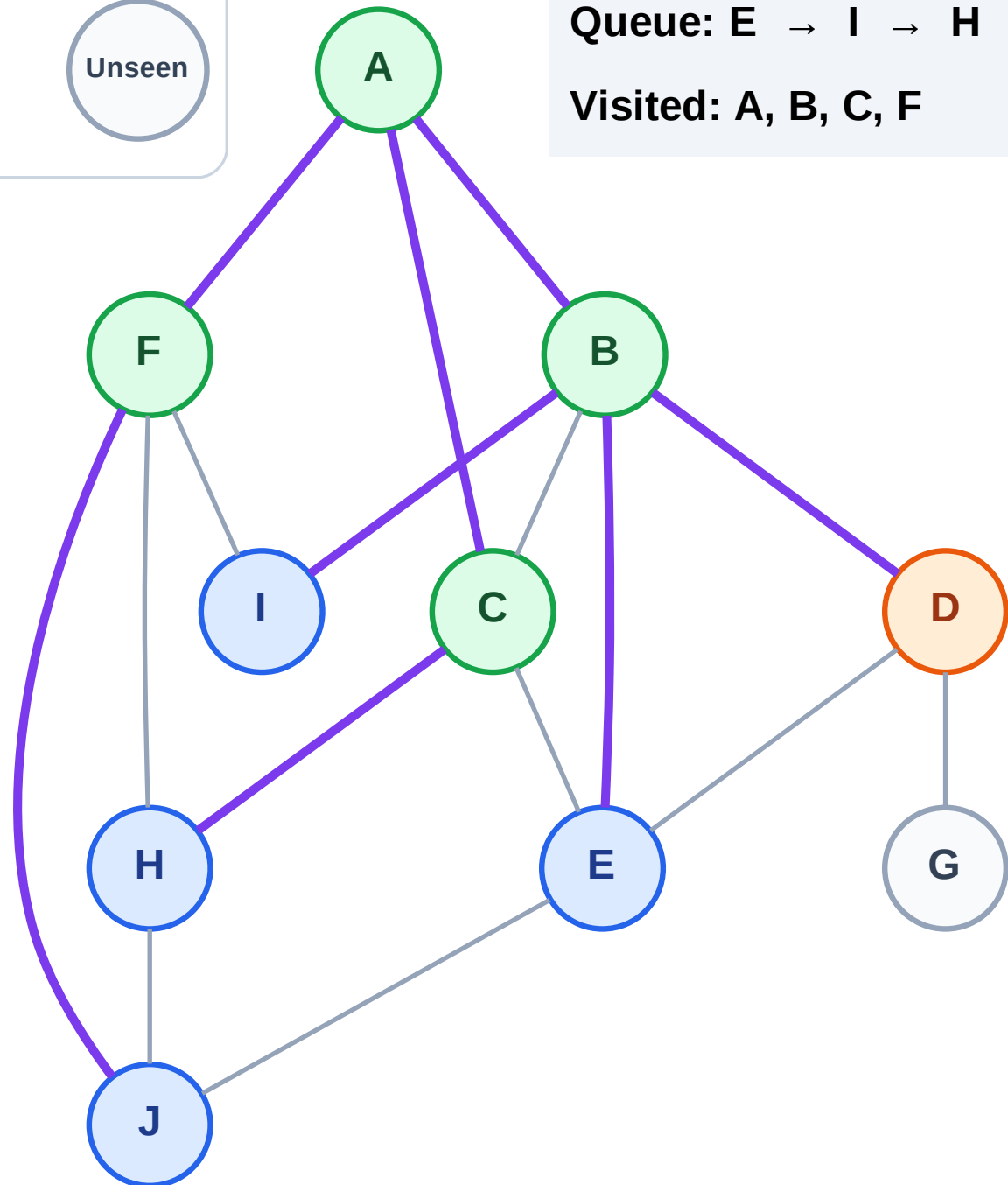
Processing

Discovered

Unseen

Queue: E → I → H → J

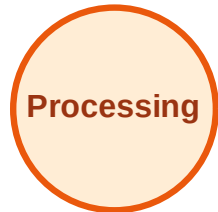
Visited: A, B, C, F



BFS Traversal

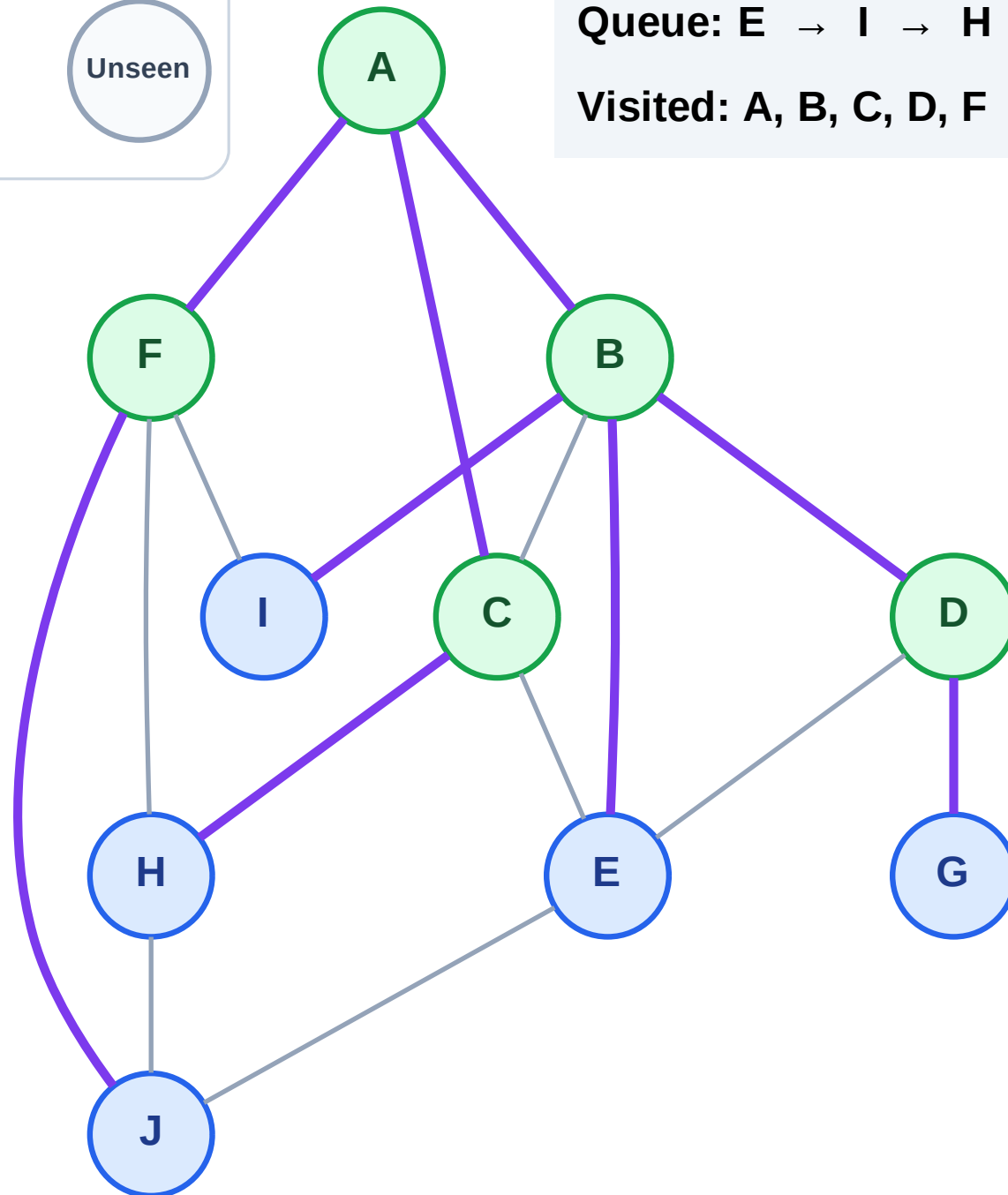
Step 11: Discover G from D

Legend



Queue: E → I → H → J → G

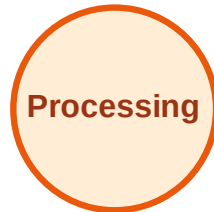
Visited: A, B, C, D, F



BFS Traversal

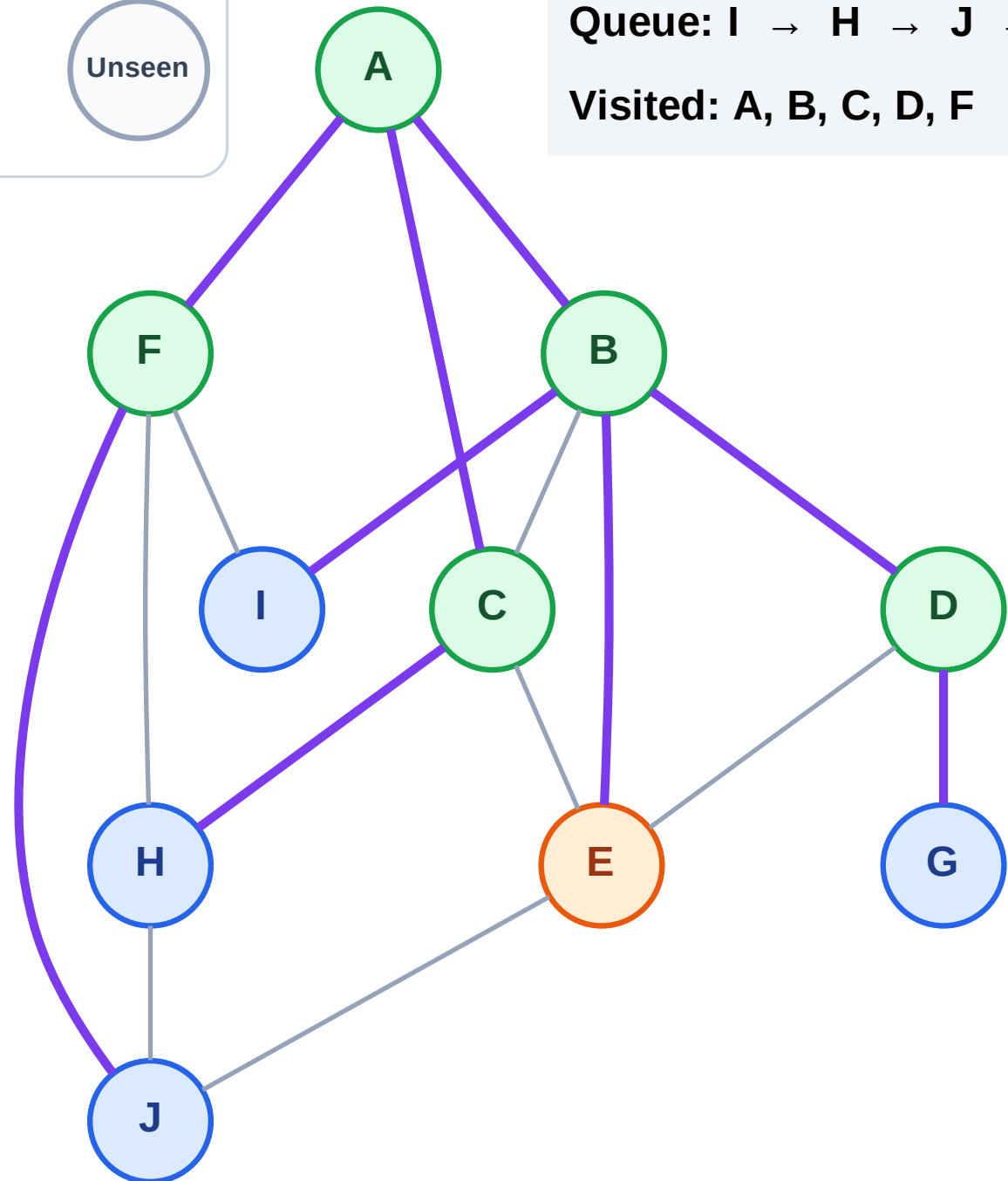
Step 12: Process node E

Legend



Queue: I → H → J → G

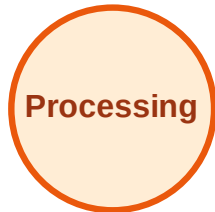
Visited: A, B, C, D, F



BFS Traversal

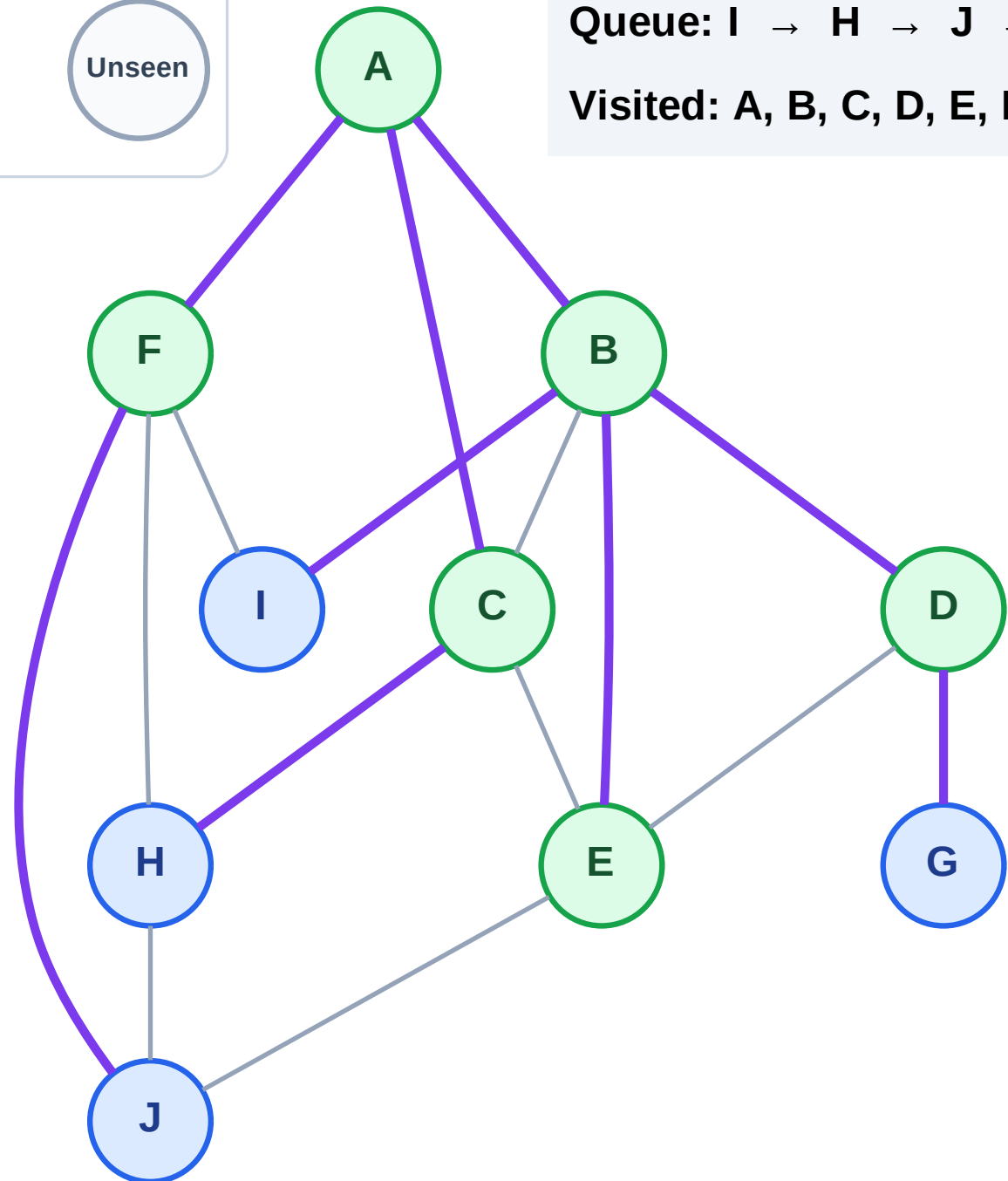
Step 13: No new neighbors from E

Legend



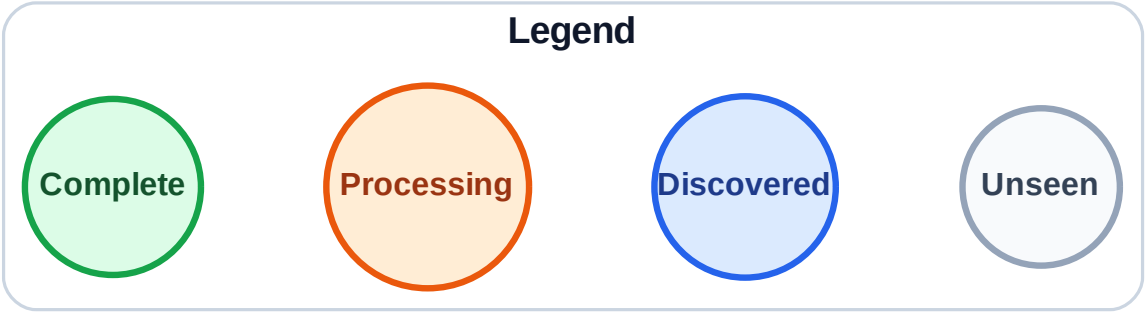
Queue: I → H → J → G

Visited: A, B, C, D, E, F



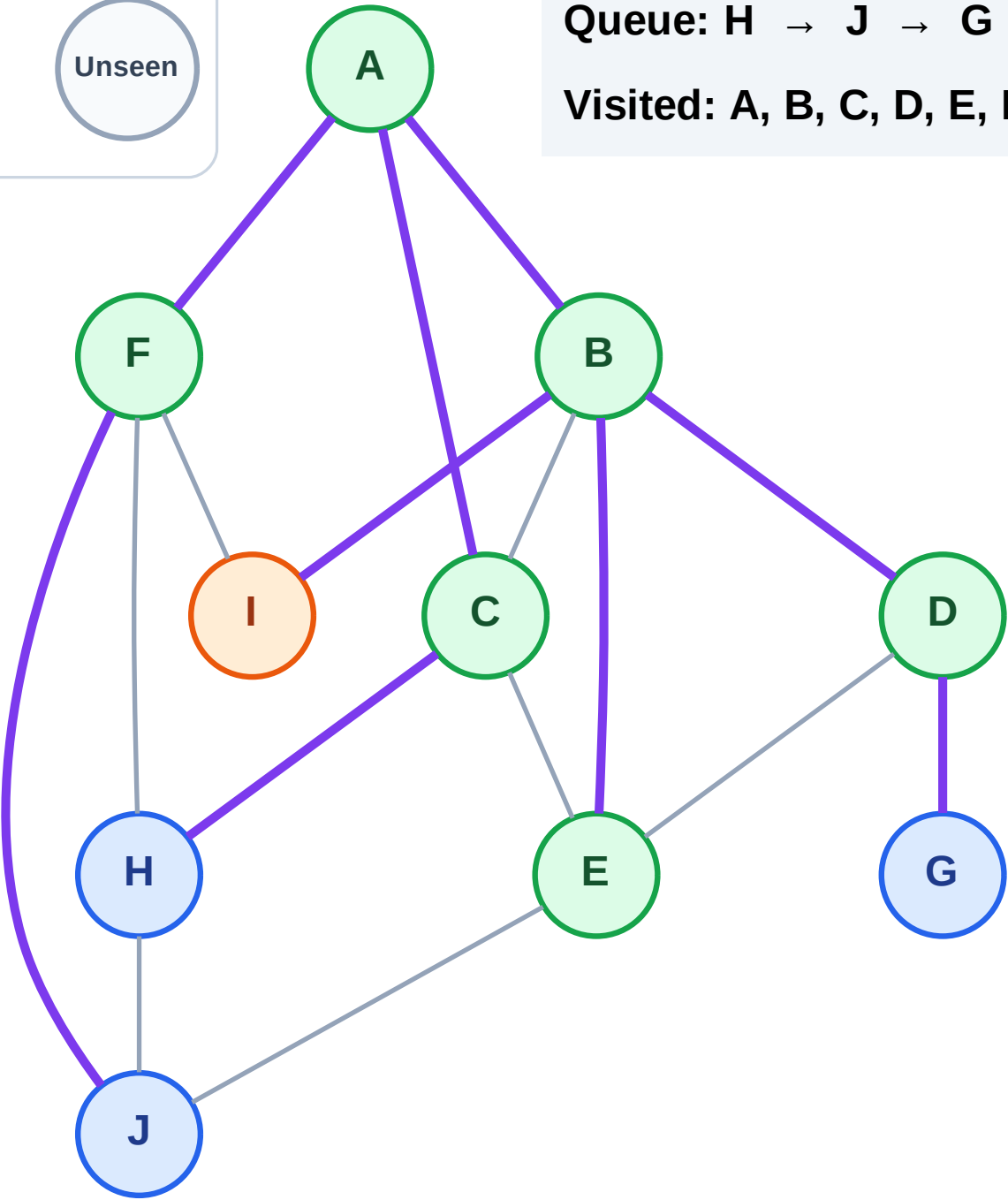
BFS Traversal

Step 14: Process node I



Queue: H → J → G

Visited: A, B, C, D, E, F



BFS Traversal

Step 15: No new neighbors from I

Legend

Complete

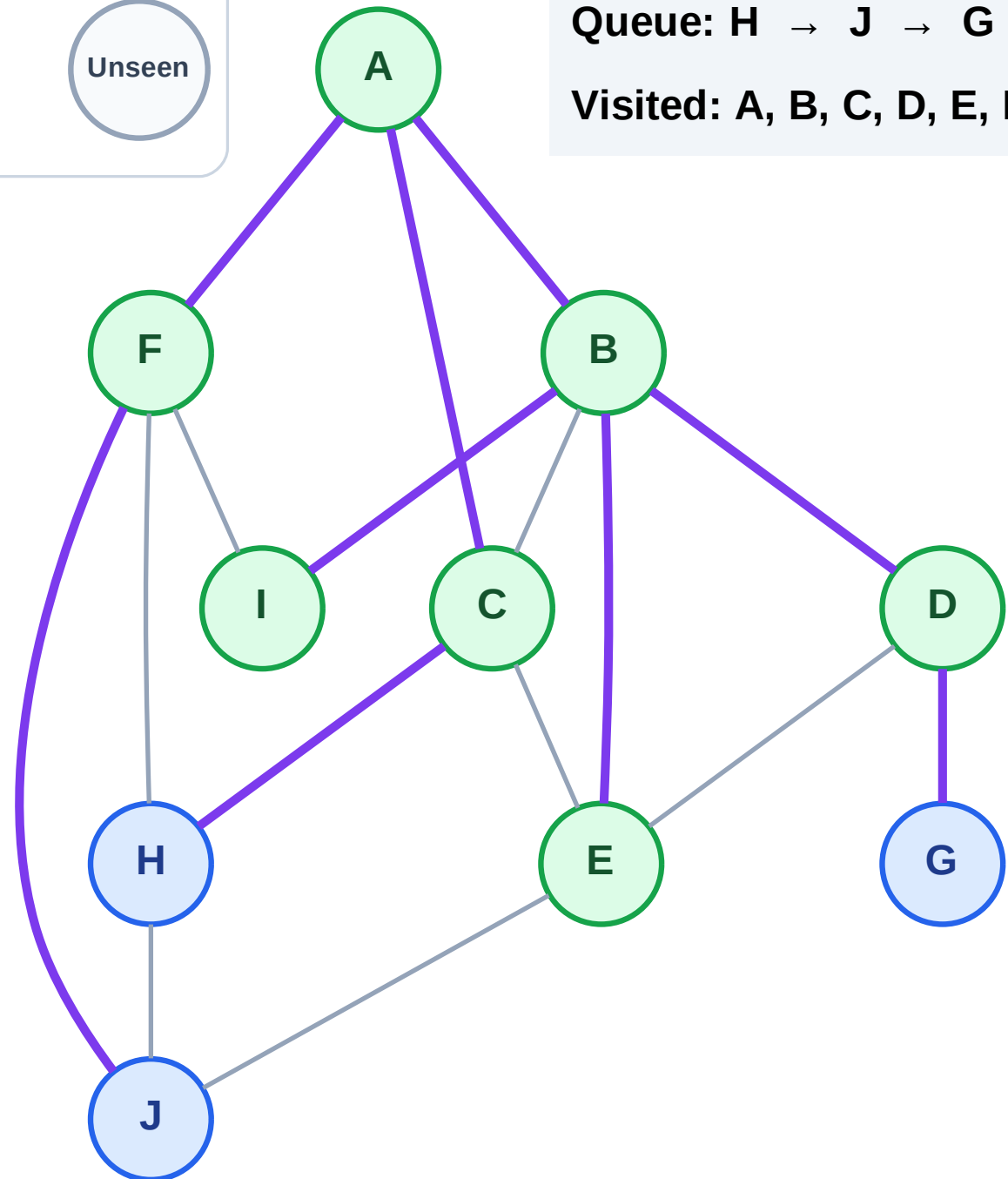
Processing

Discovered

Unseen

Queue: H → J → G

Visited: A, B, C, D, E, F, I



BFS Traversal

Step 16: Process node H

Legend

Complete

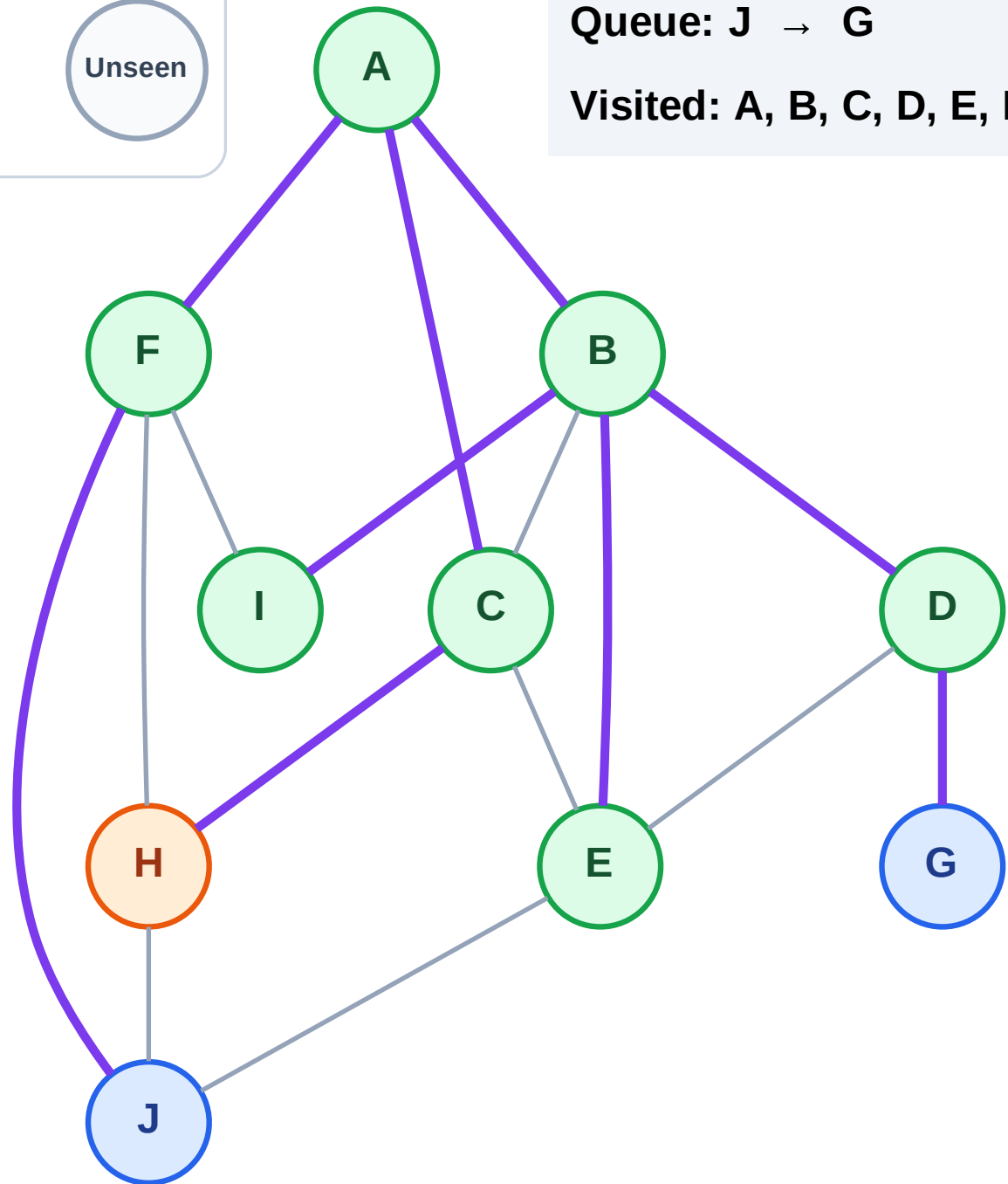
Processing

Discovered

Unseen

Queue: J → G

Visited: A, B, C, D, E, F, I



BFS Traversal

Step 17: No new neighbors from H

Legend

Complete

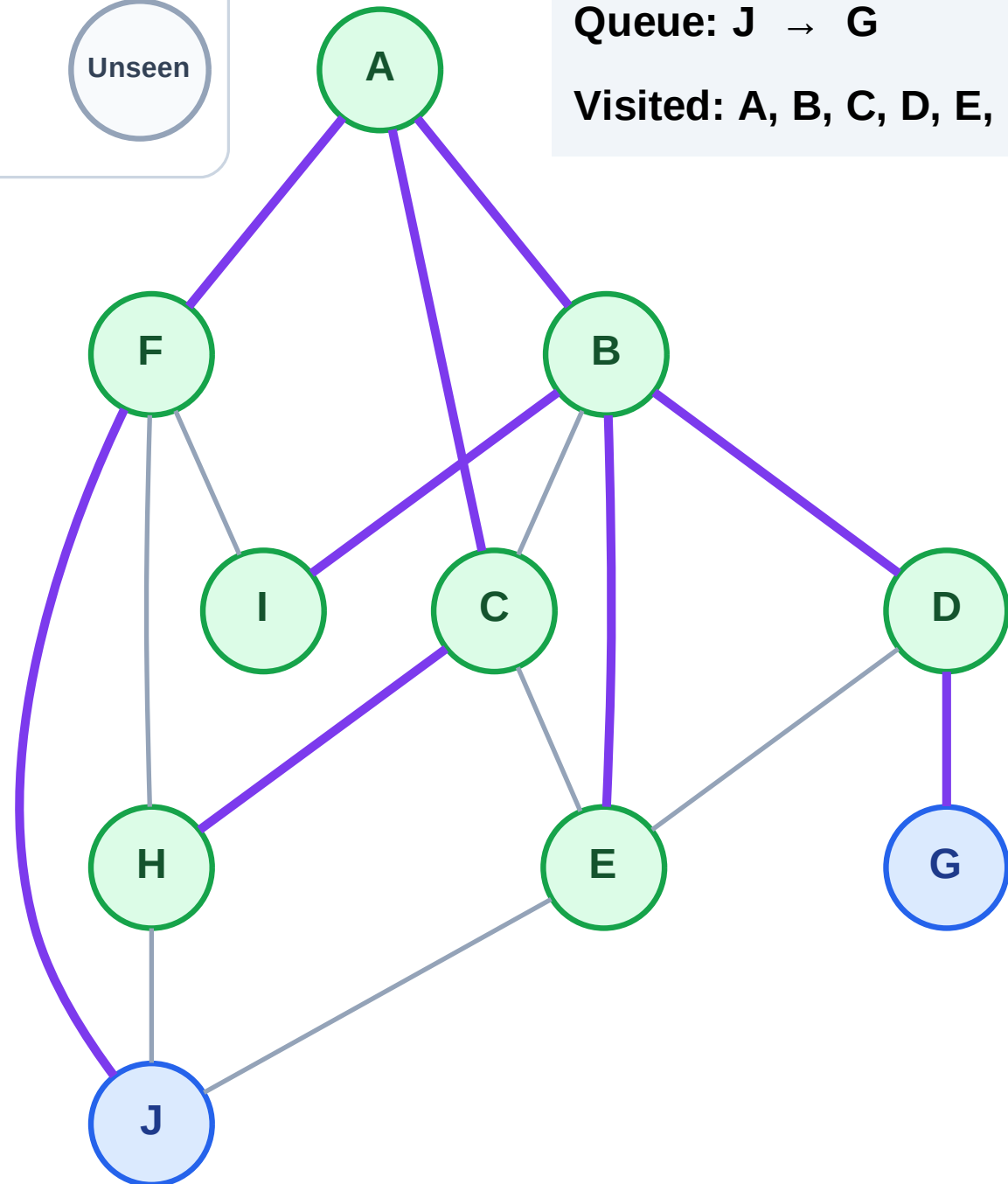
Processing

Discovered

Unseen

Queue: J → G

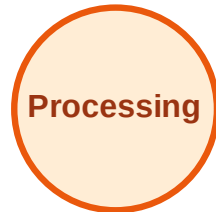
Visited: A, B, C, D, E, F, H, I



BFS Traversal

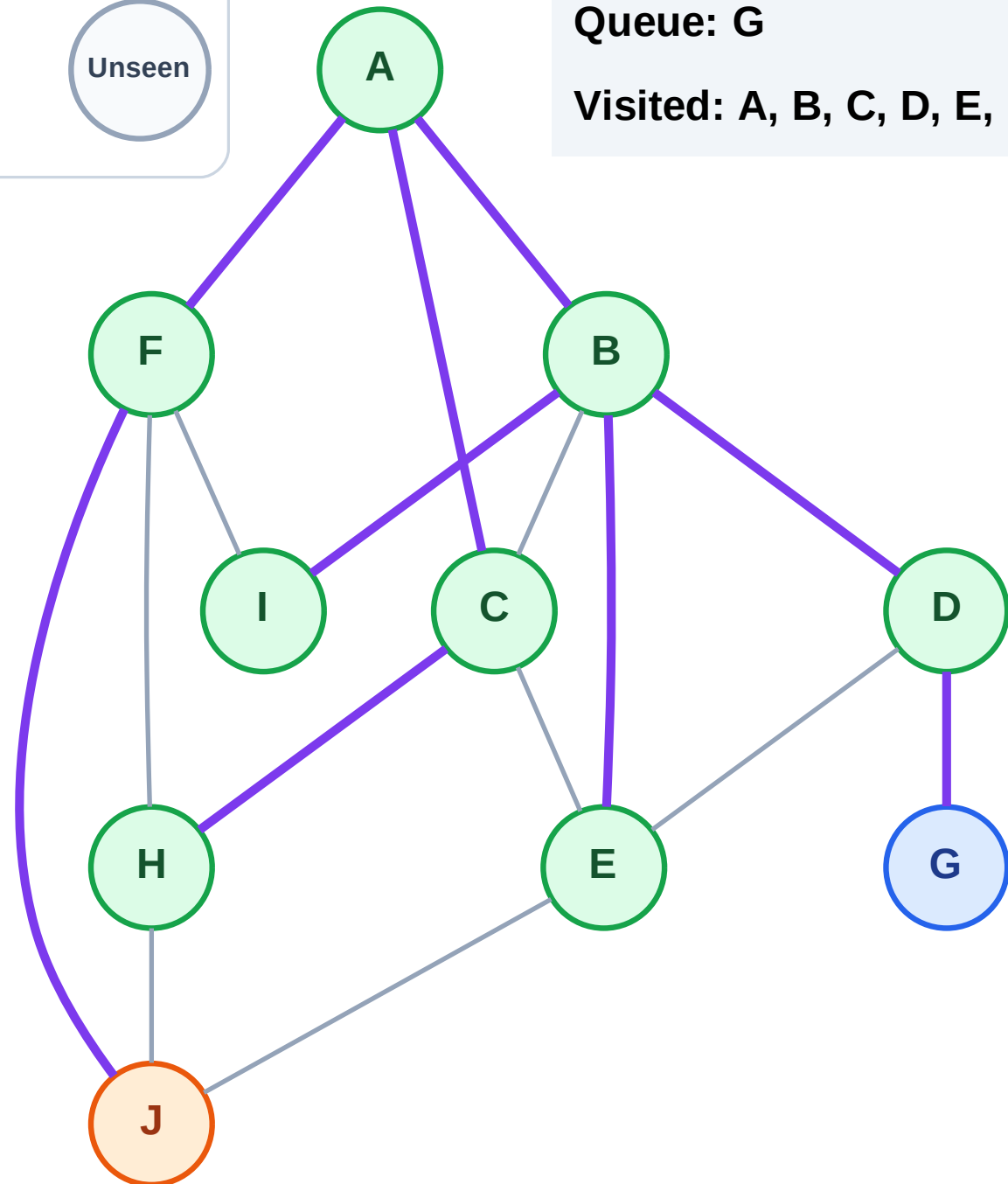
Step 18: Process node J

Legend



Queue: G

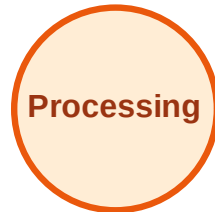
Visited: A, B, C, D, E, F, H, I



BFS Traversal

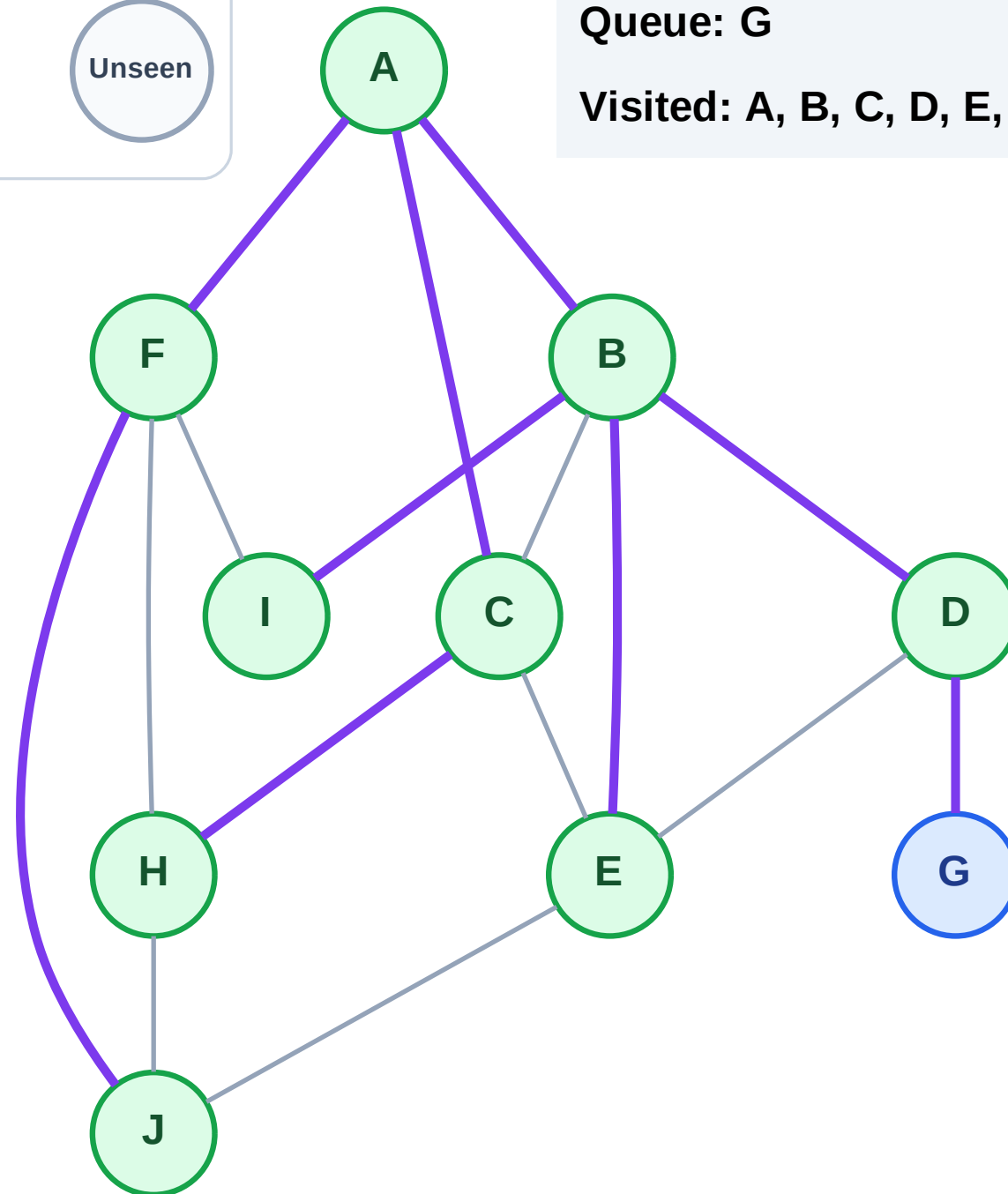
Step 19: No new neighbors from J

Legend



Queue: G

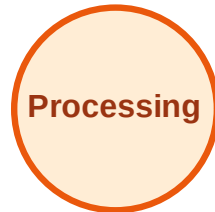
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

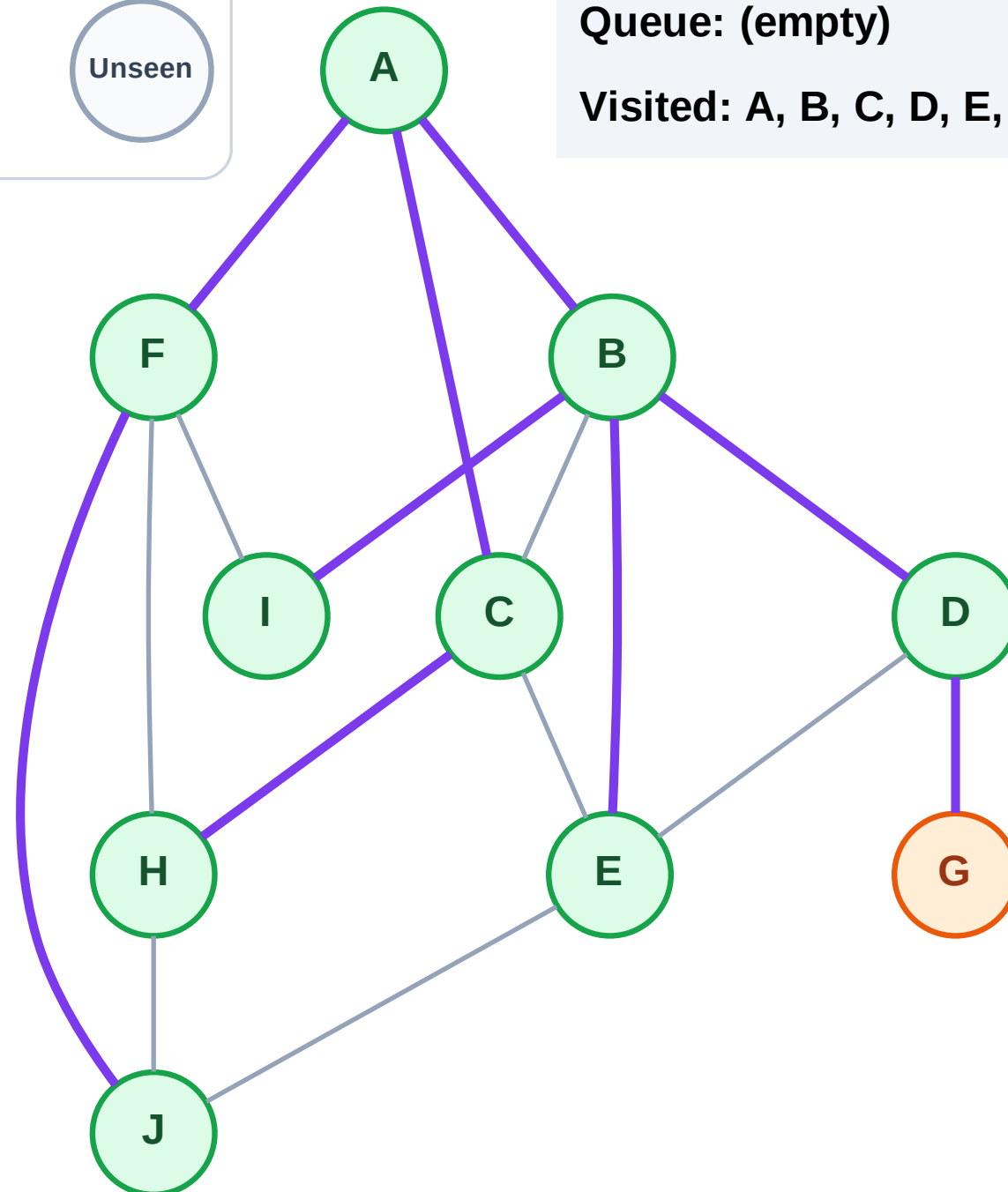
Step 20: Process node G

Legend



Queue: (empty)

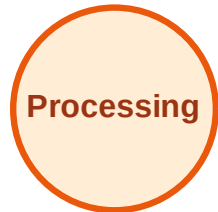
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

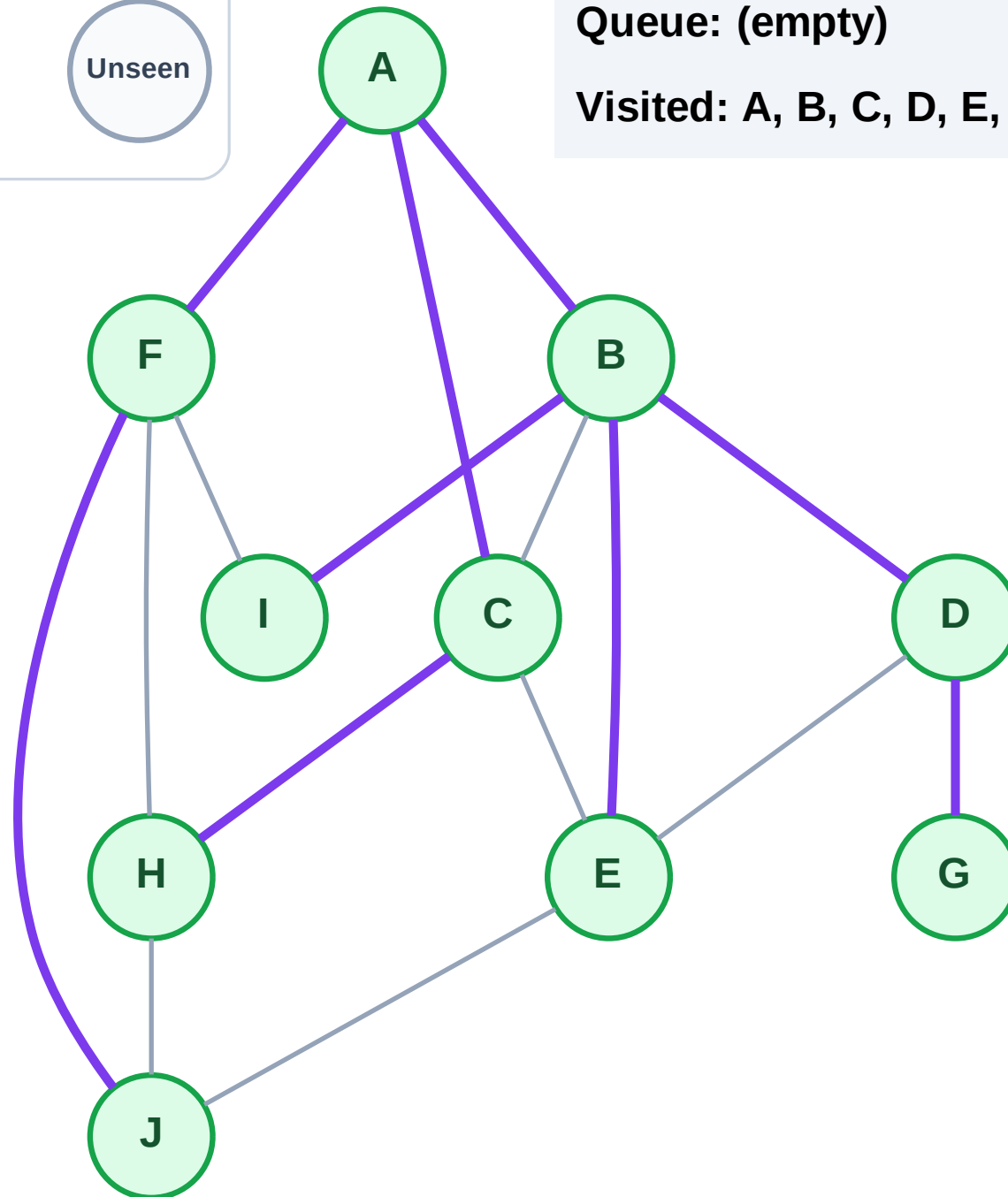
Step 21: No new neighbors from G

Legend



Queue: (empty)

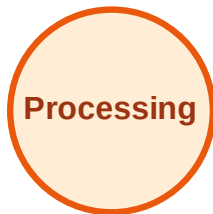
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

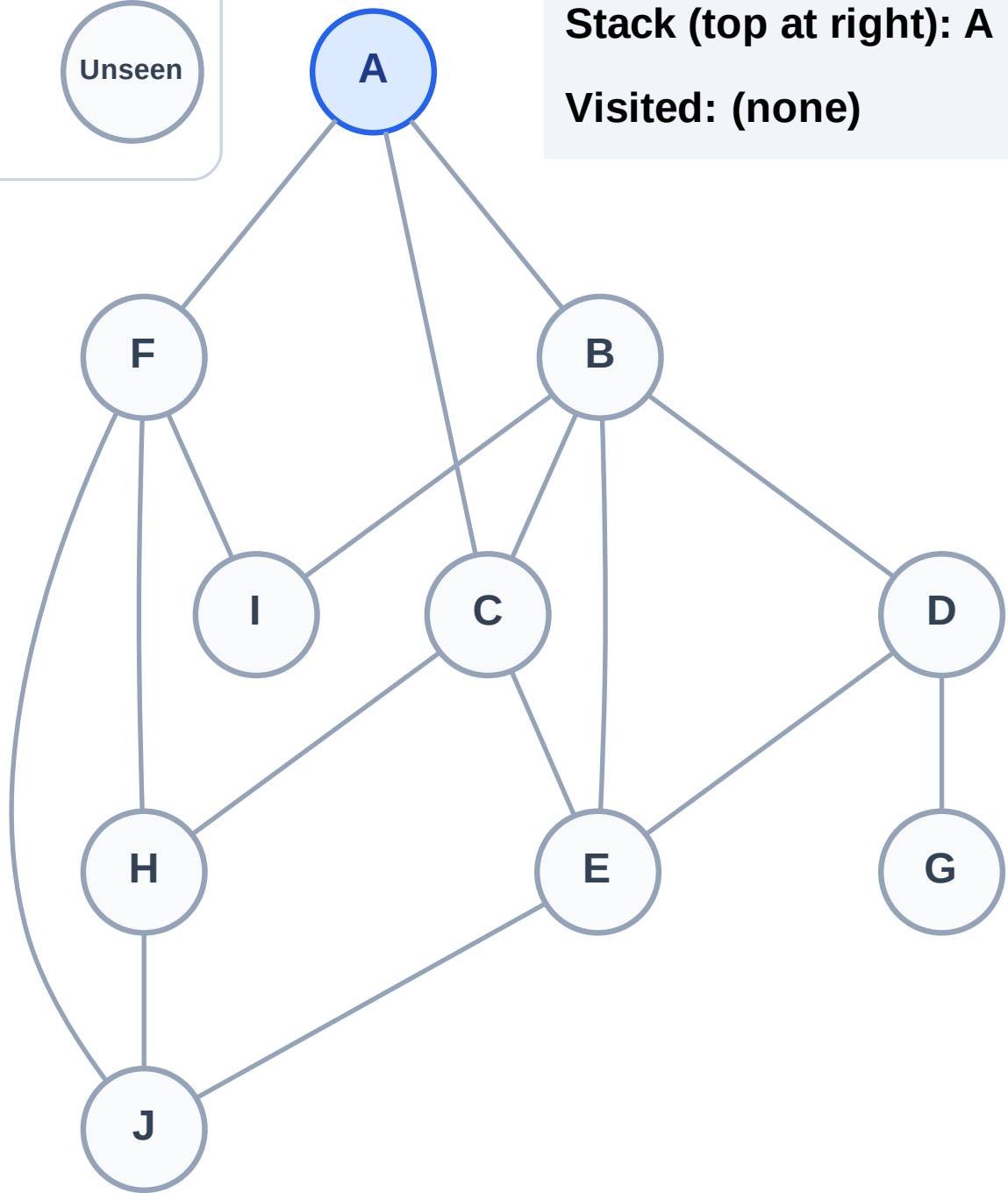
Step 1: Start at A

Legend



Stack (top at right): A

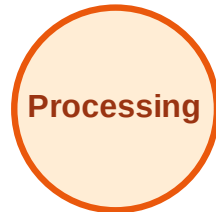
Visited: (none)



DFS Traversal

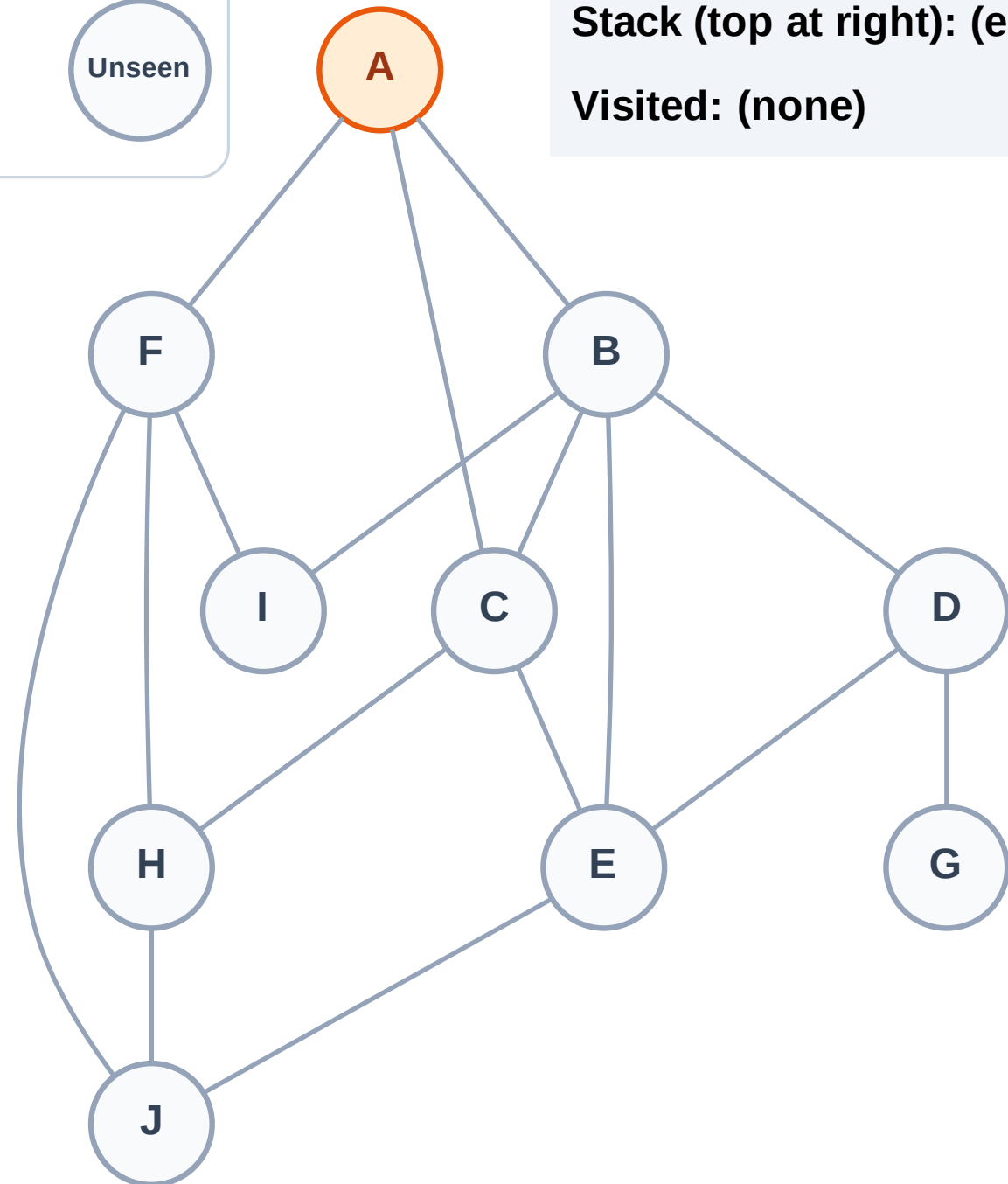
Step 2: Process node A

Legend



Stack (top at right): (empty)

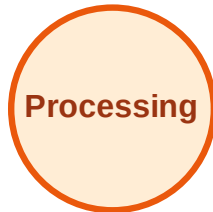
Visited: (none)



DFS Traversal

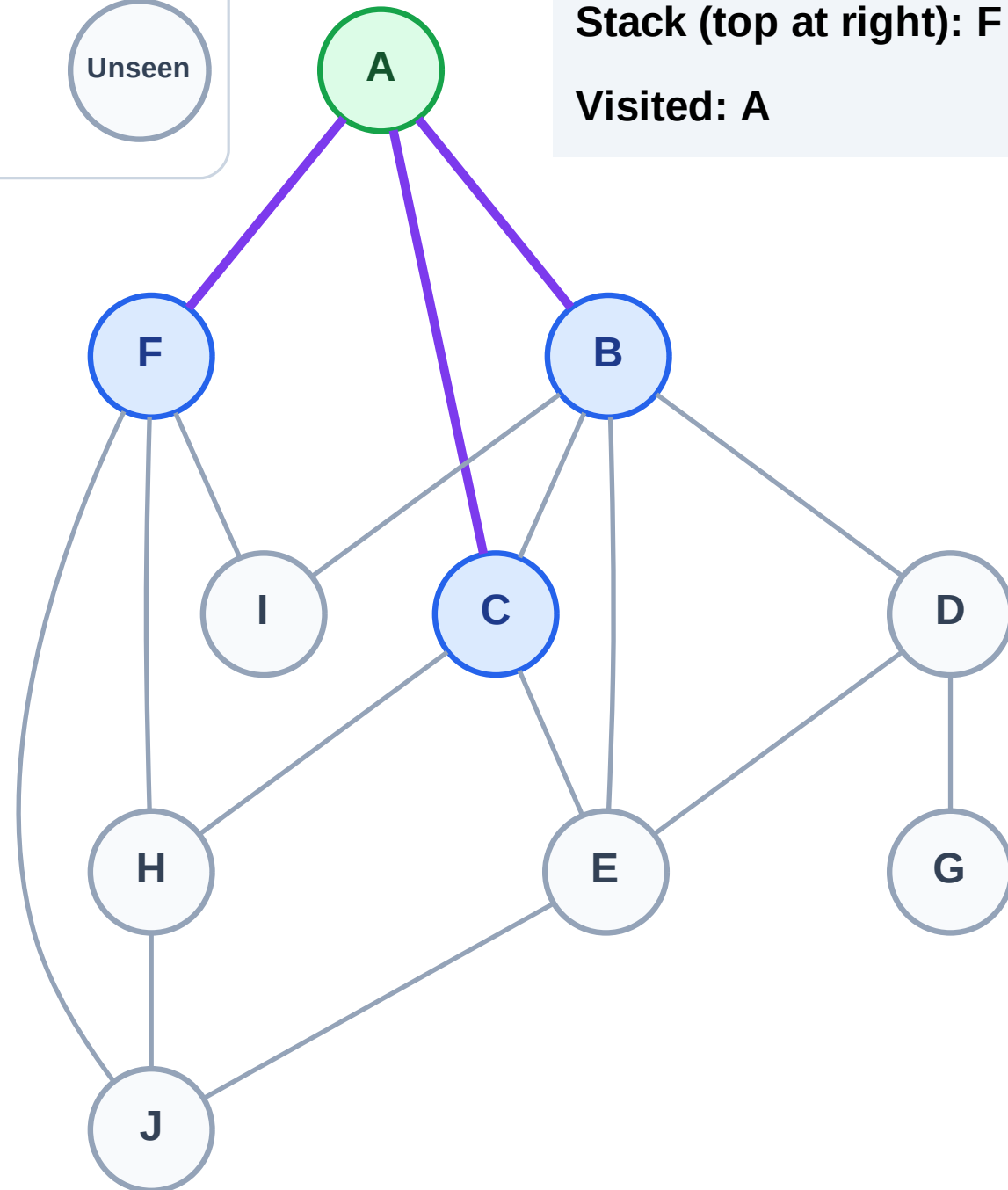
Step 3: Discover F, C, B from A

Legend



Stack (top at right): F | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

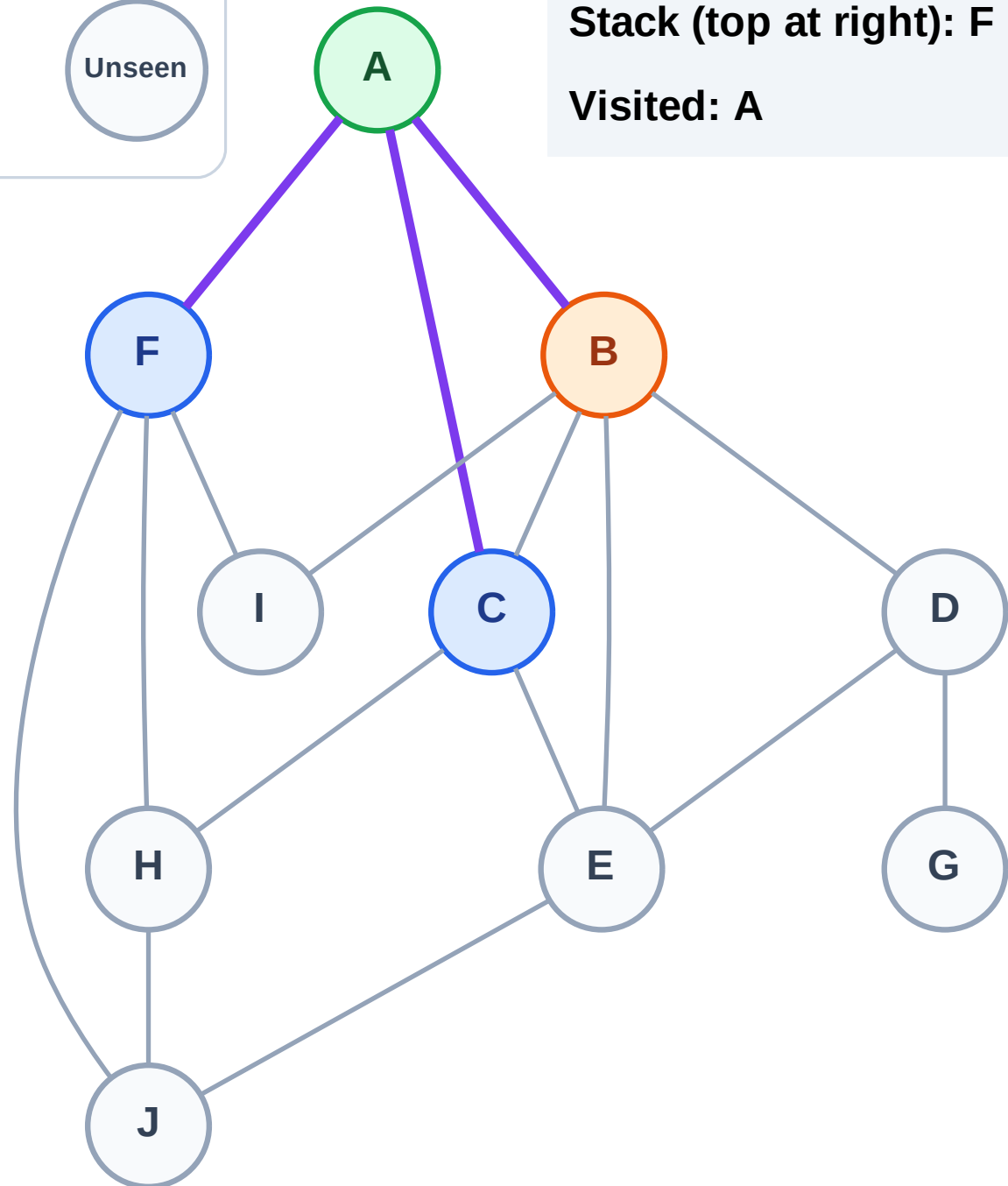
Processing

Discovered

Unseen

Stack (top at right): F | C

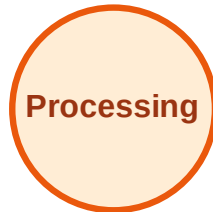
Visited: A



DFS Traversal

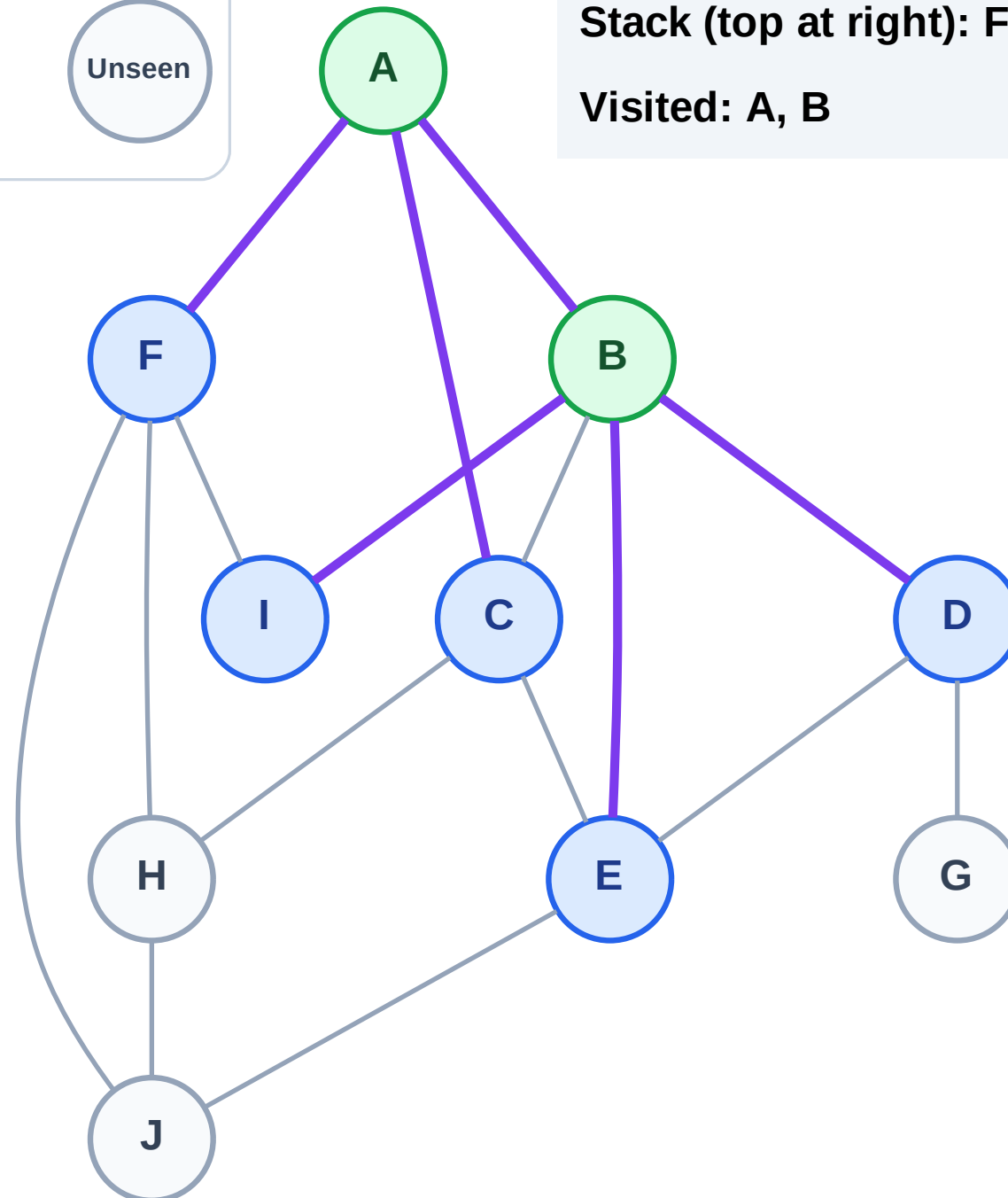
Step 5: Discover I, E, D from B

Legend



Stack (top at right): F | C | I | E | D

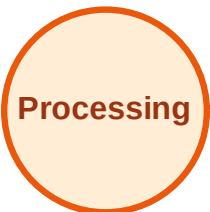
Visited: A, B



DFS Traversal

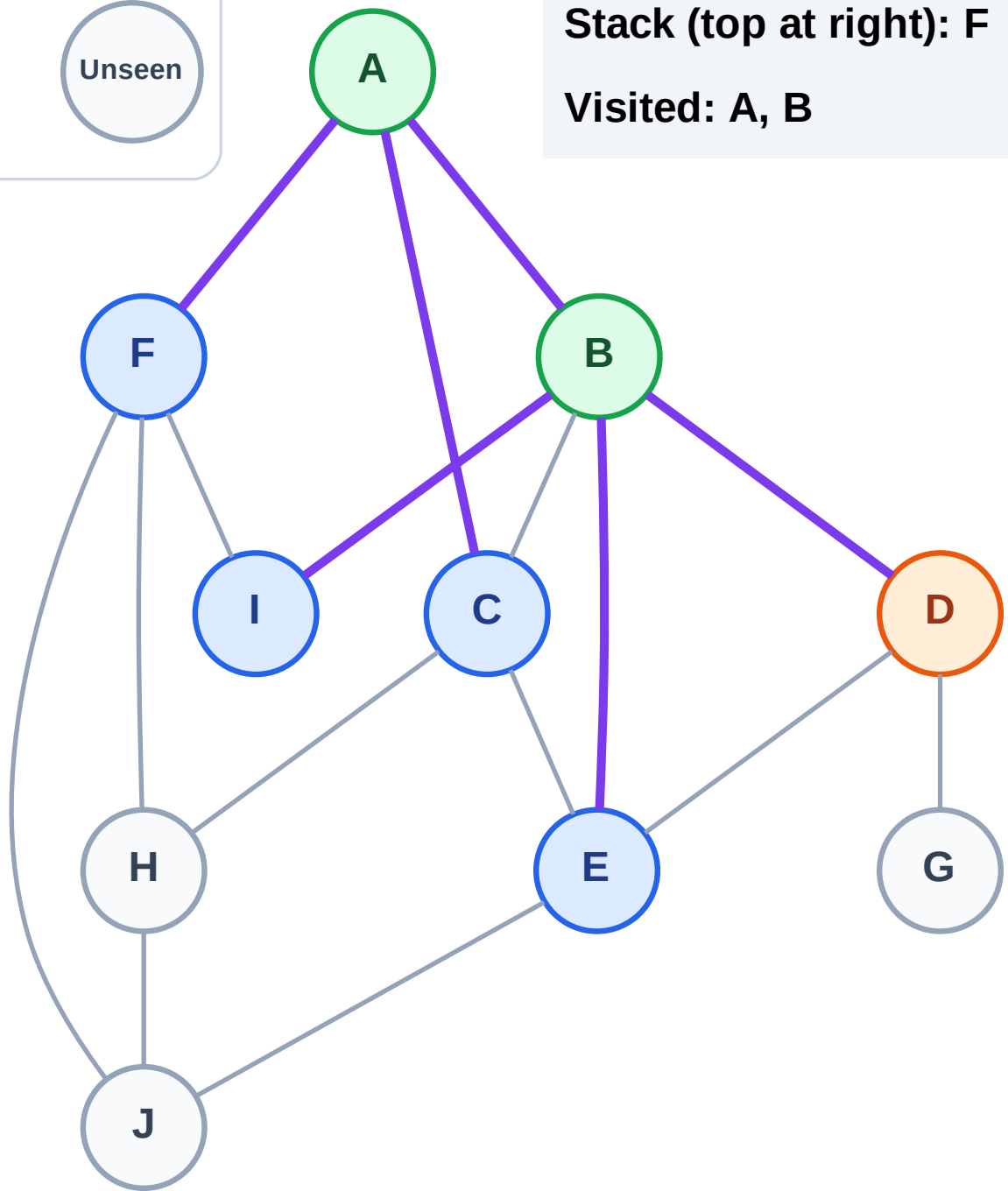
Step 6: Process node D

Legend



Stack (top at right): F | C | I | E

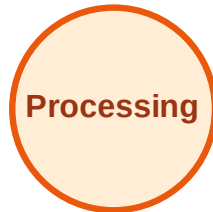
Visited: A, B



DFS Traversal

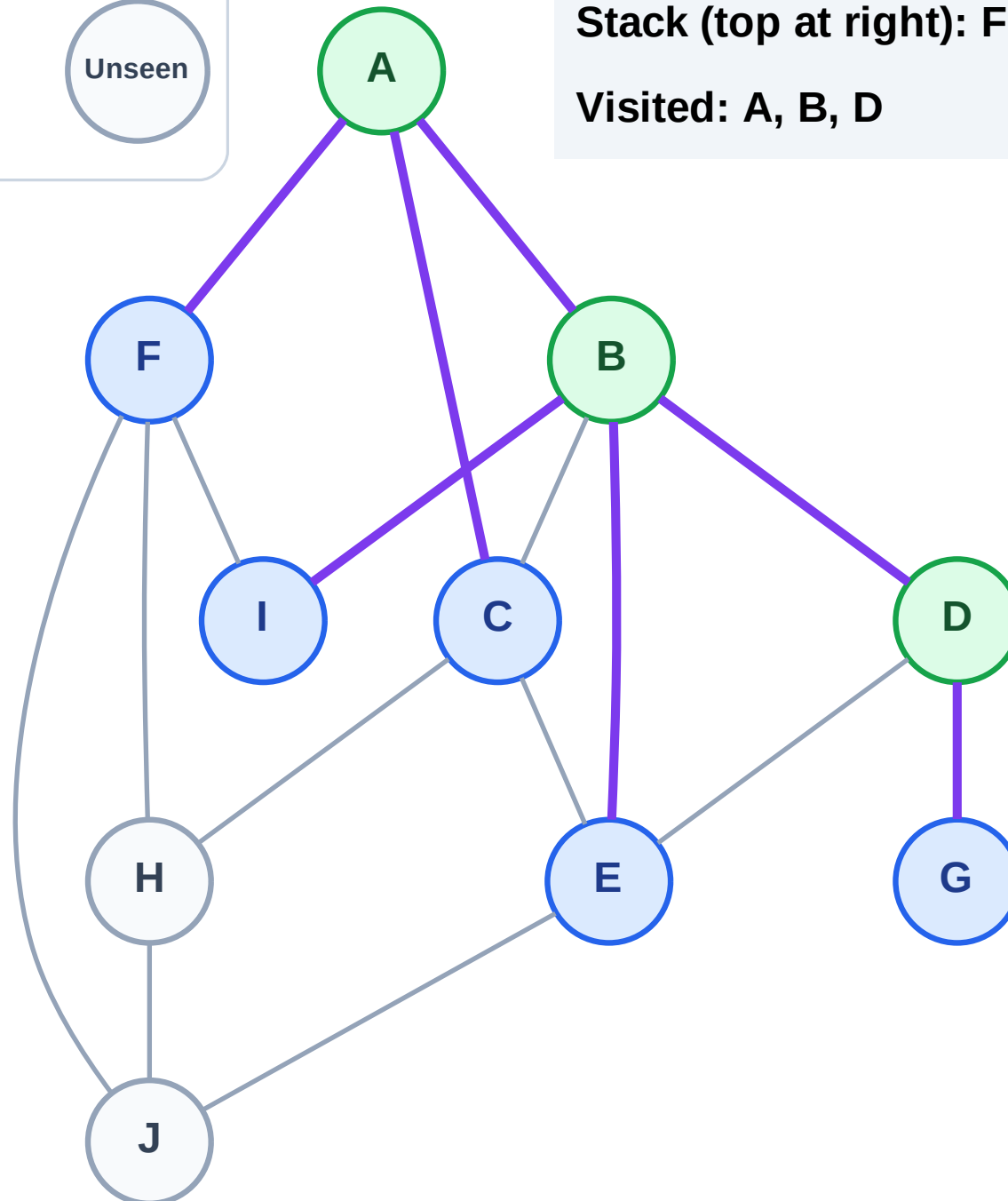
Step 7: Discover G from D

Legend



Stack (top at right): F | C | I | E | G

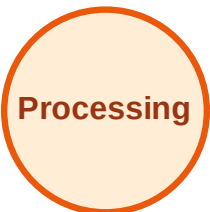
Visited: A, B, D



DFS Traversal

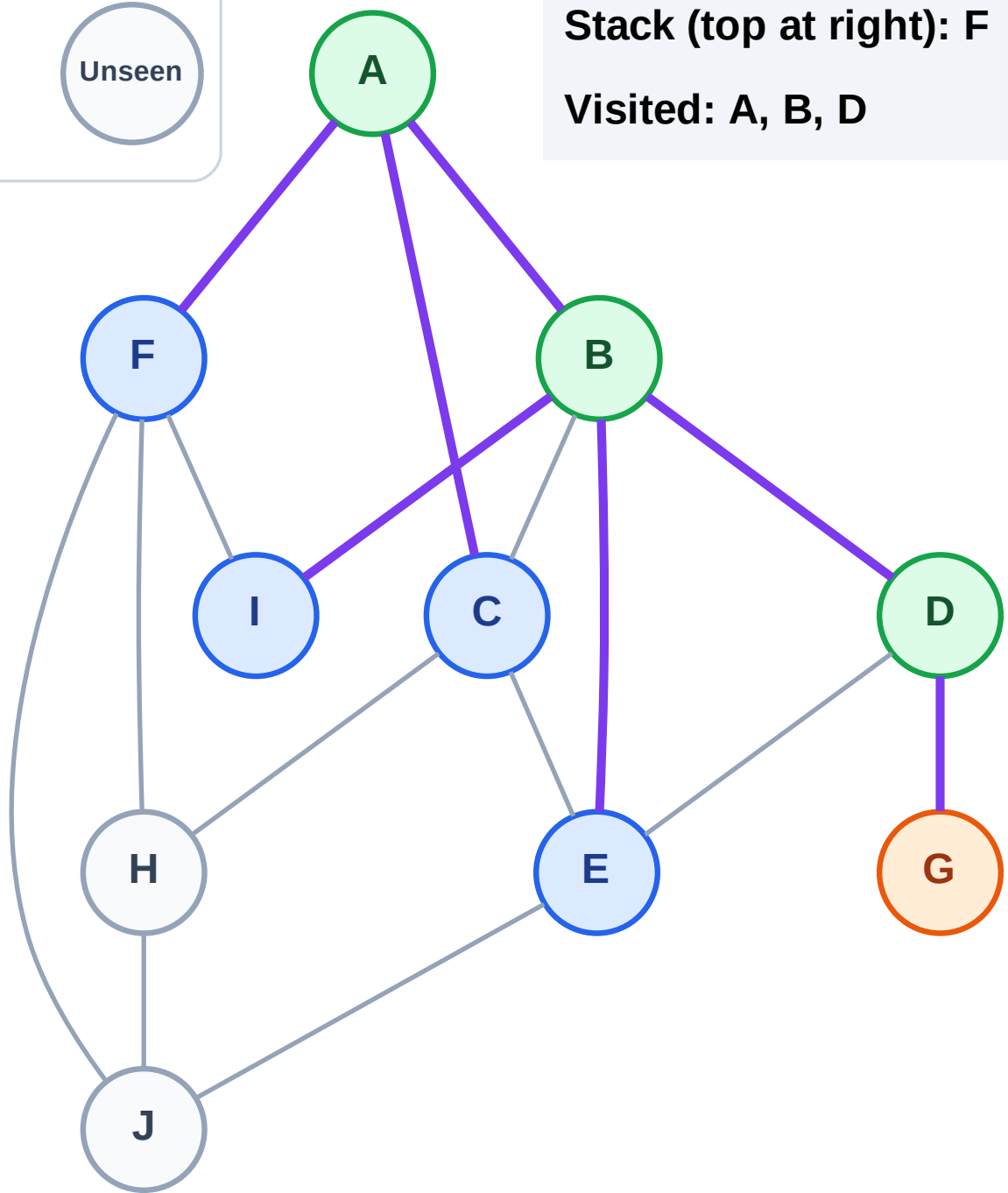
Step 8: Process node G

Legend



Stack (top at right): F | C | I | E

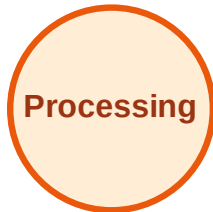
Visited: A, B, D



DFS Traversal

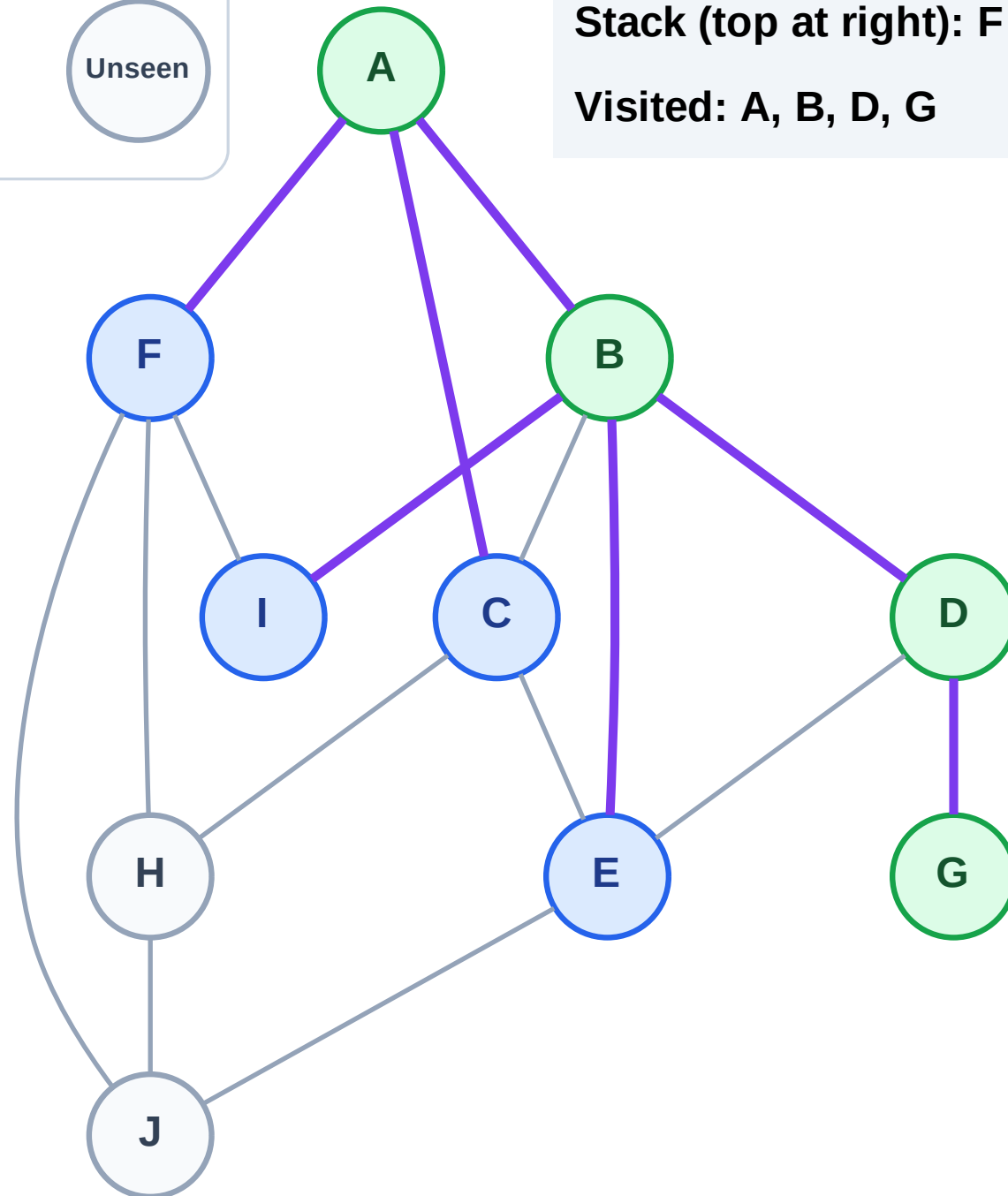
Step 9: No new neighbors from G

Legend



Stack (top at right): F | C | I | E

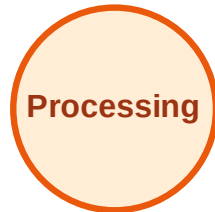
Visited: A, B, D, G



DFS Traversal

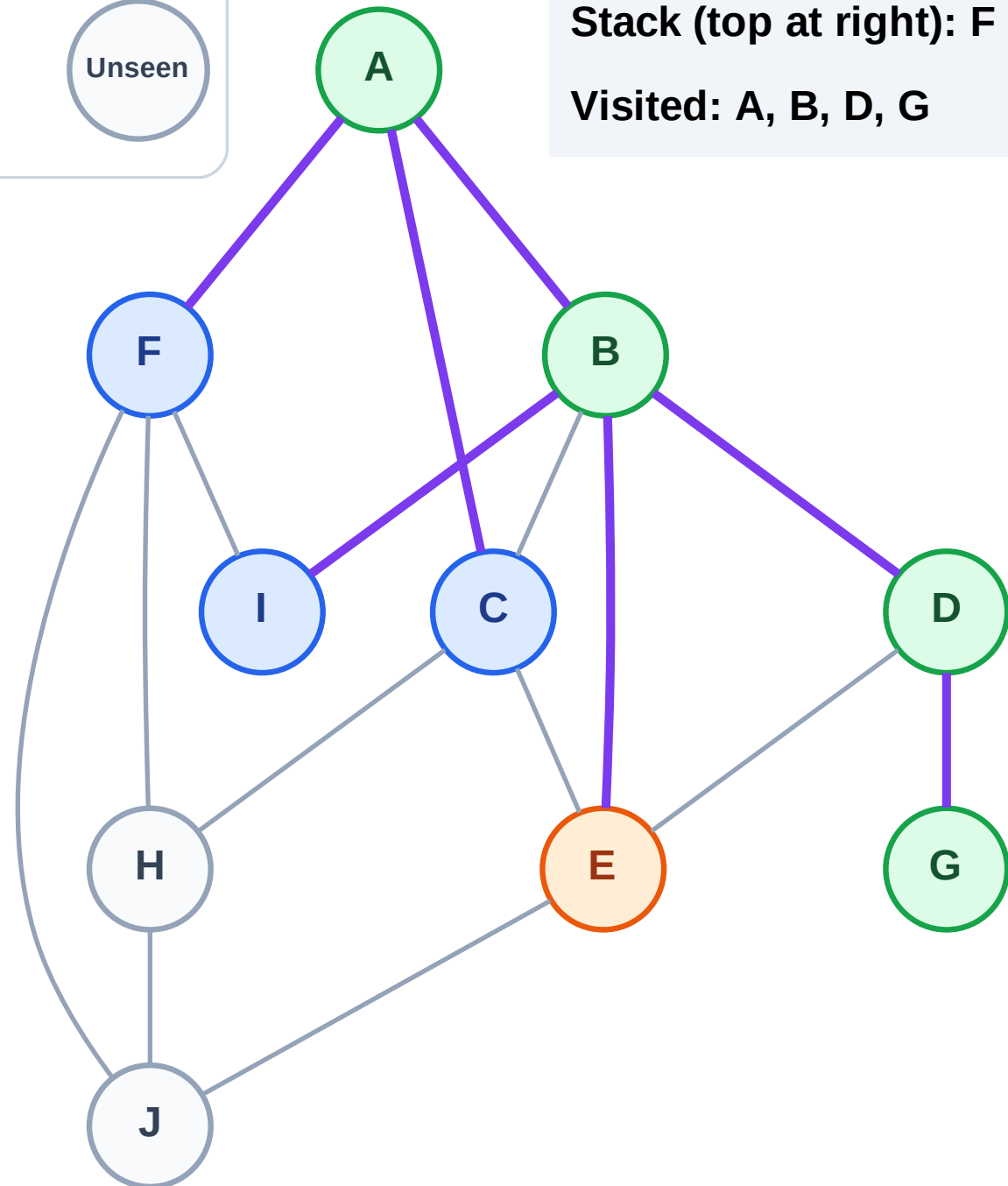
Step 10: Process node E

Legend



Stack (top at right): F | C | I

Visited: A, B, D, G



DFS Traversal

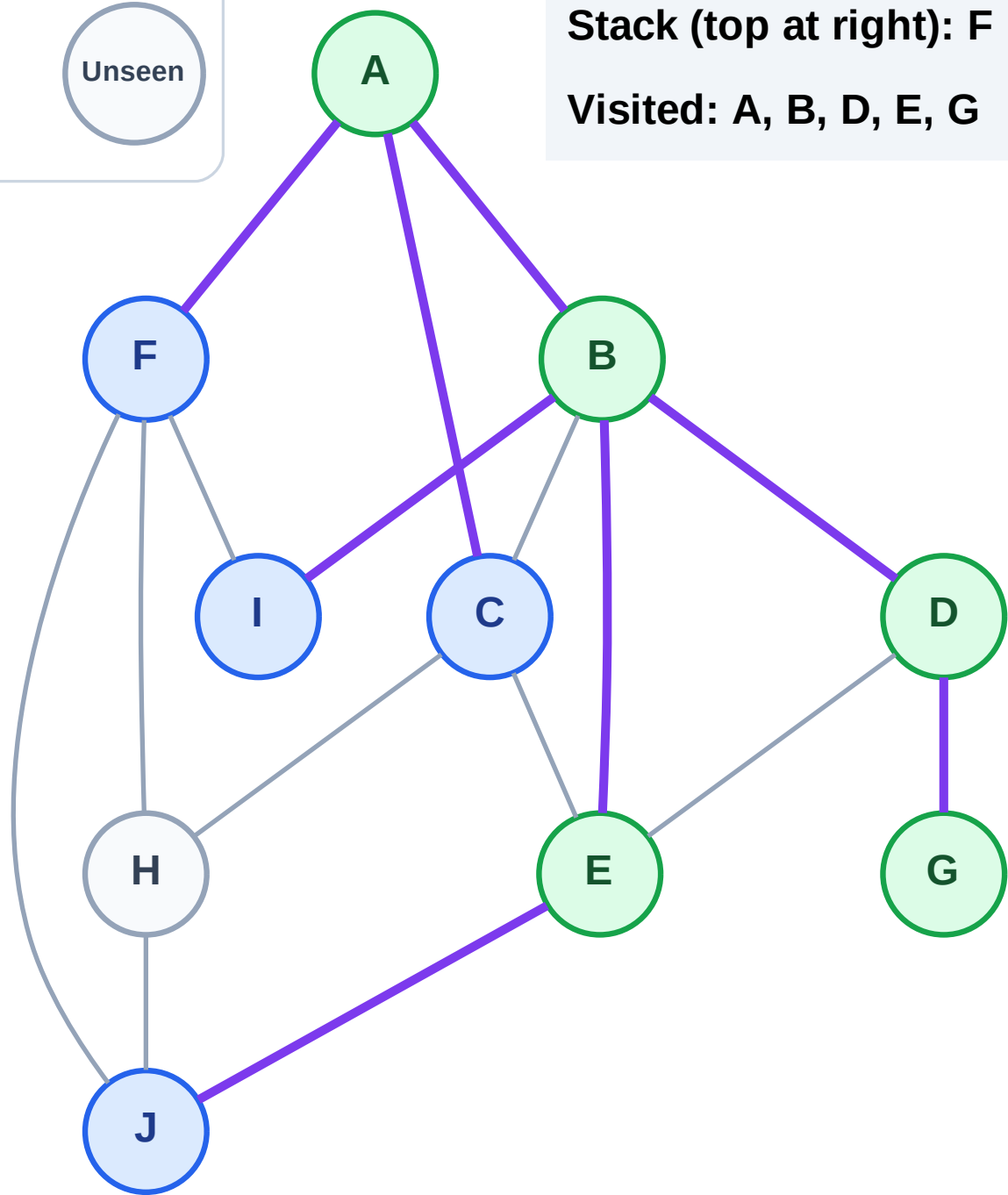
Step 11: Discover J from E

Legend



Stack (top at right): F | C | I | J

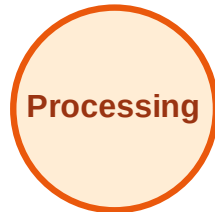
Visited: A, B, D, E, G



DFS Traversal

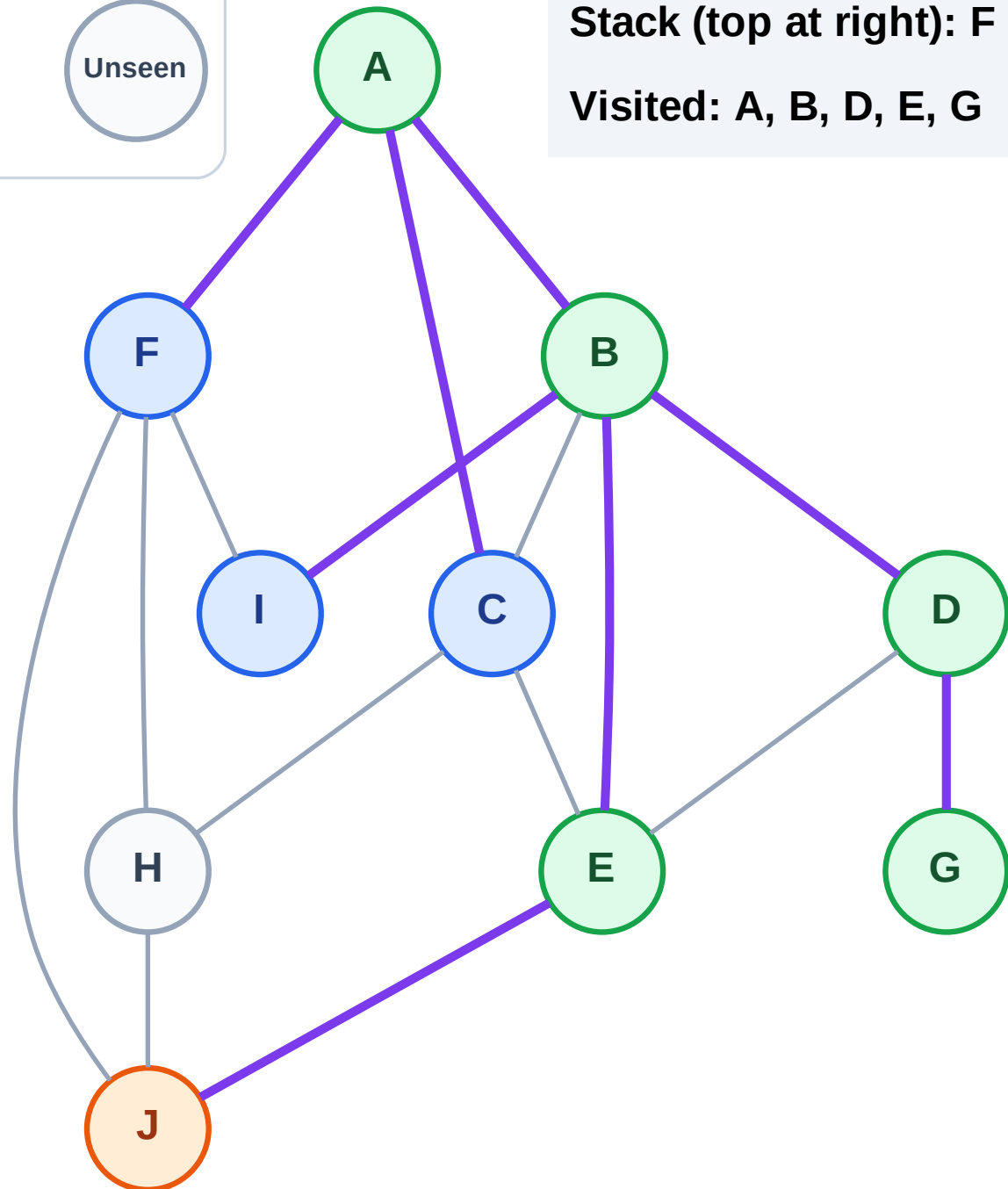
Step 12: Process node J

Legend



Stack (top at right): F | C | I

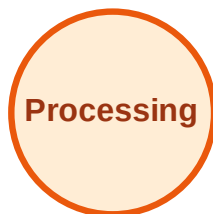
Visited: A, B, D, E, G



DFS Traversal

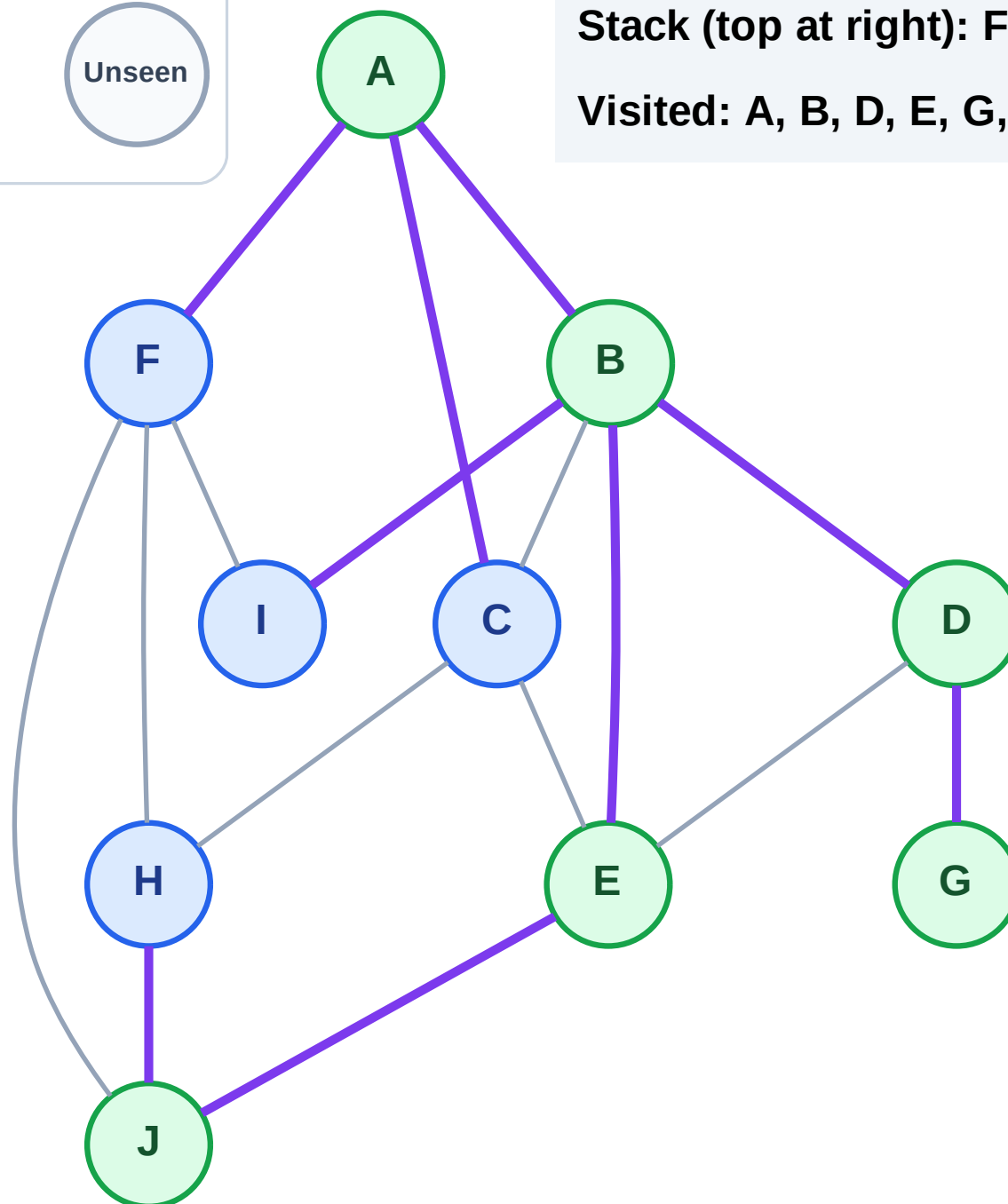
Step 13: Discover H from J

Legend



Stack (top at right): F | C | I | H

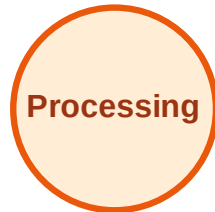
Visited: A, B, D, E, G, J



DFS Traversal

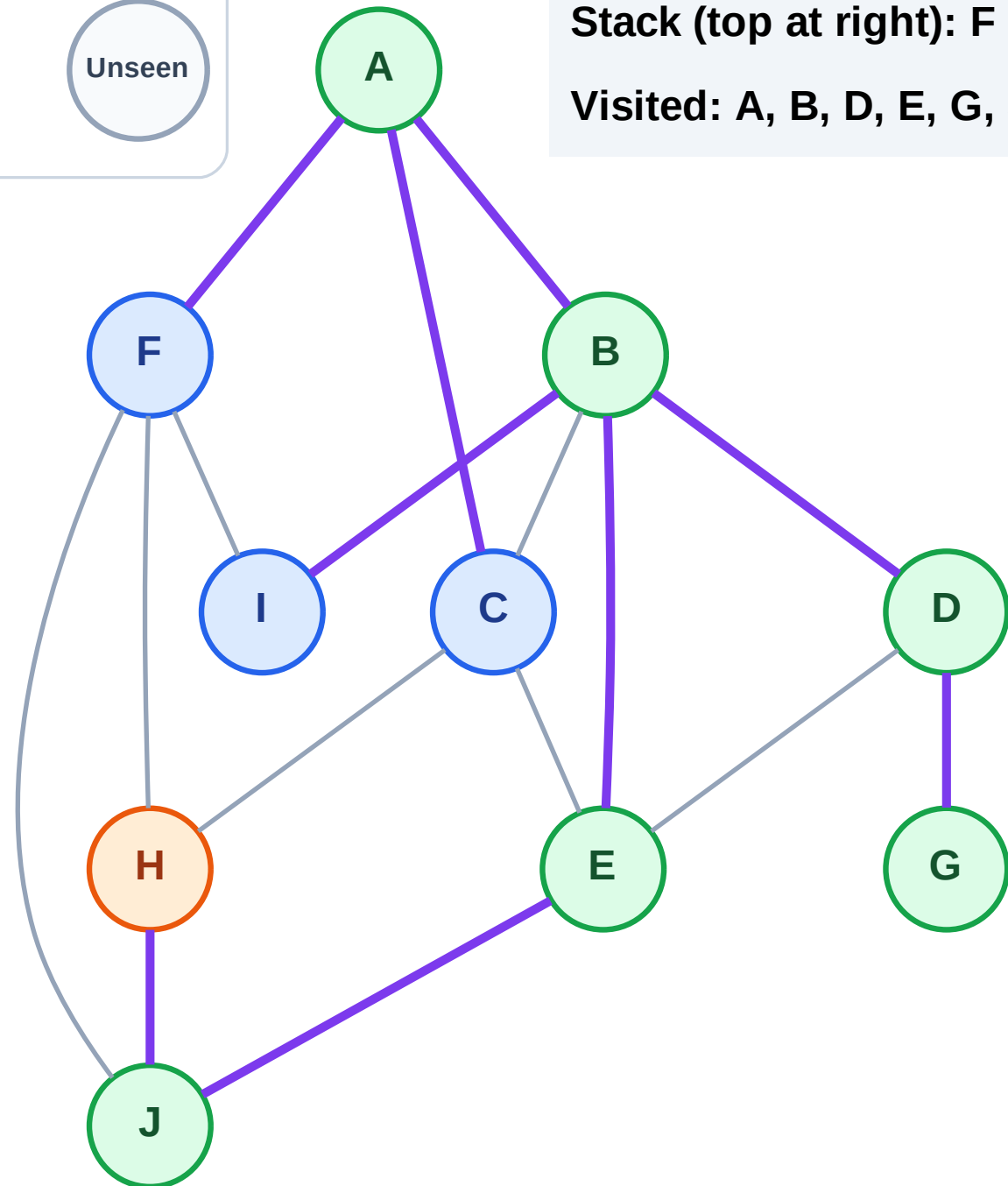
Step 14: Process node H

Legend



Stack (top at right): F | C | I

Visited: A, B, D, E, G, J

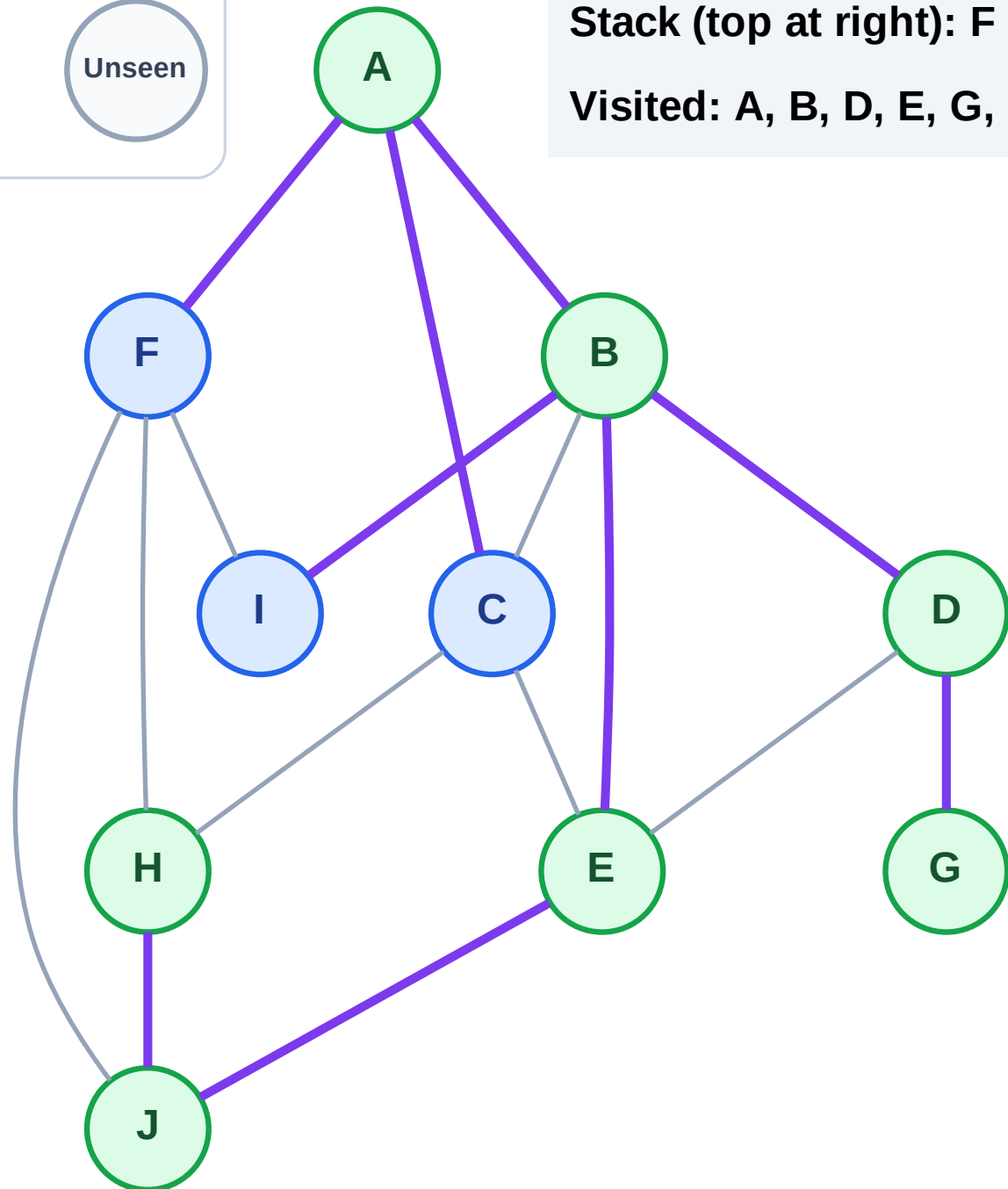


Step 15: No new neighbors from H

Legend



Visited: A, B, D, E, G, H, J



DFS Traversal

Step 16: Process node I

Legend

Complete

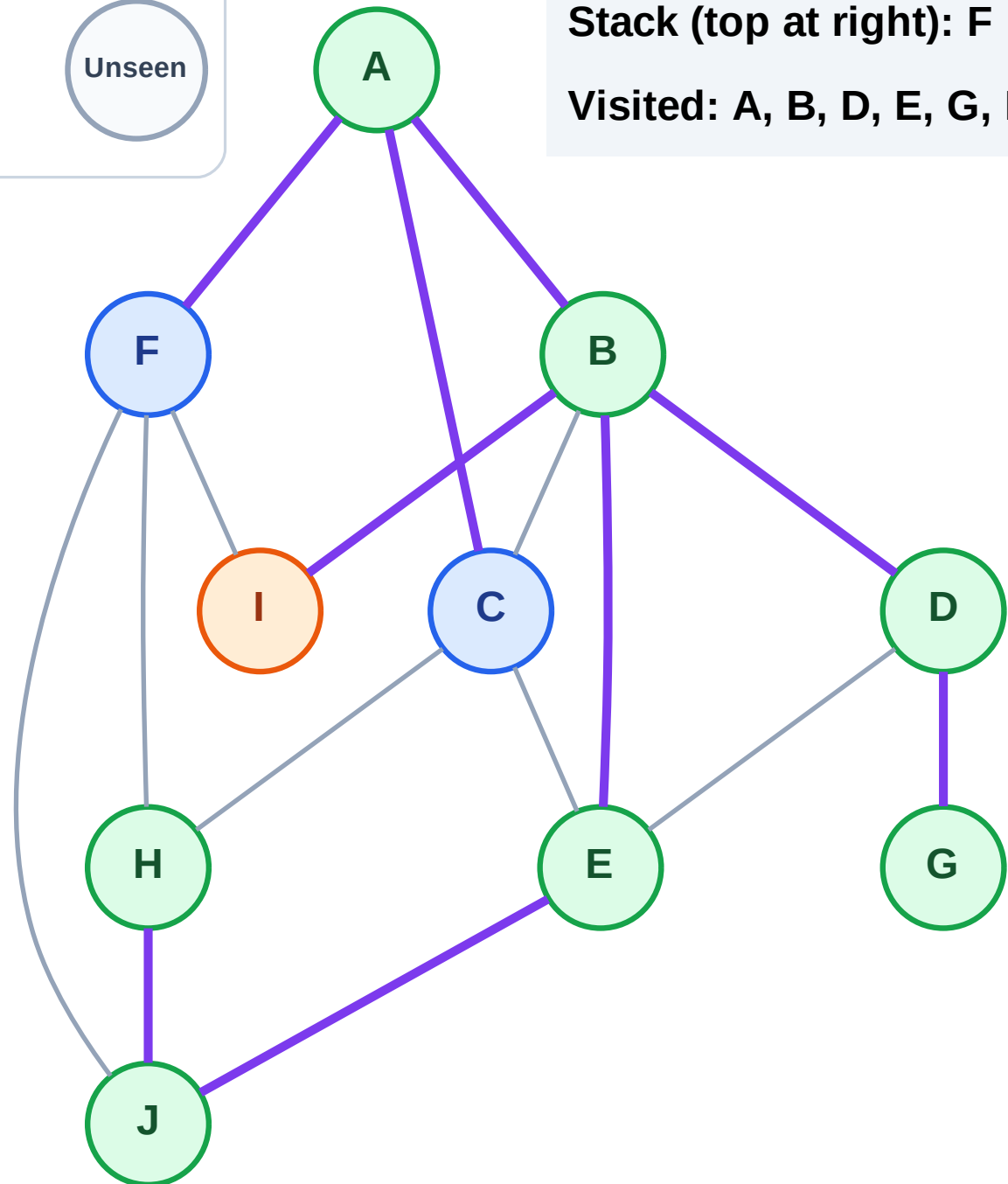
Processing

Discovered

Unseen

Stack (top at right): F | C

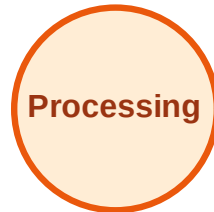
Visited: A, B, D, E, G, H, J



DFS Traversal

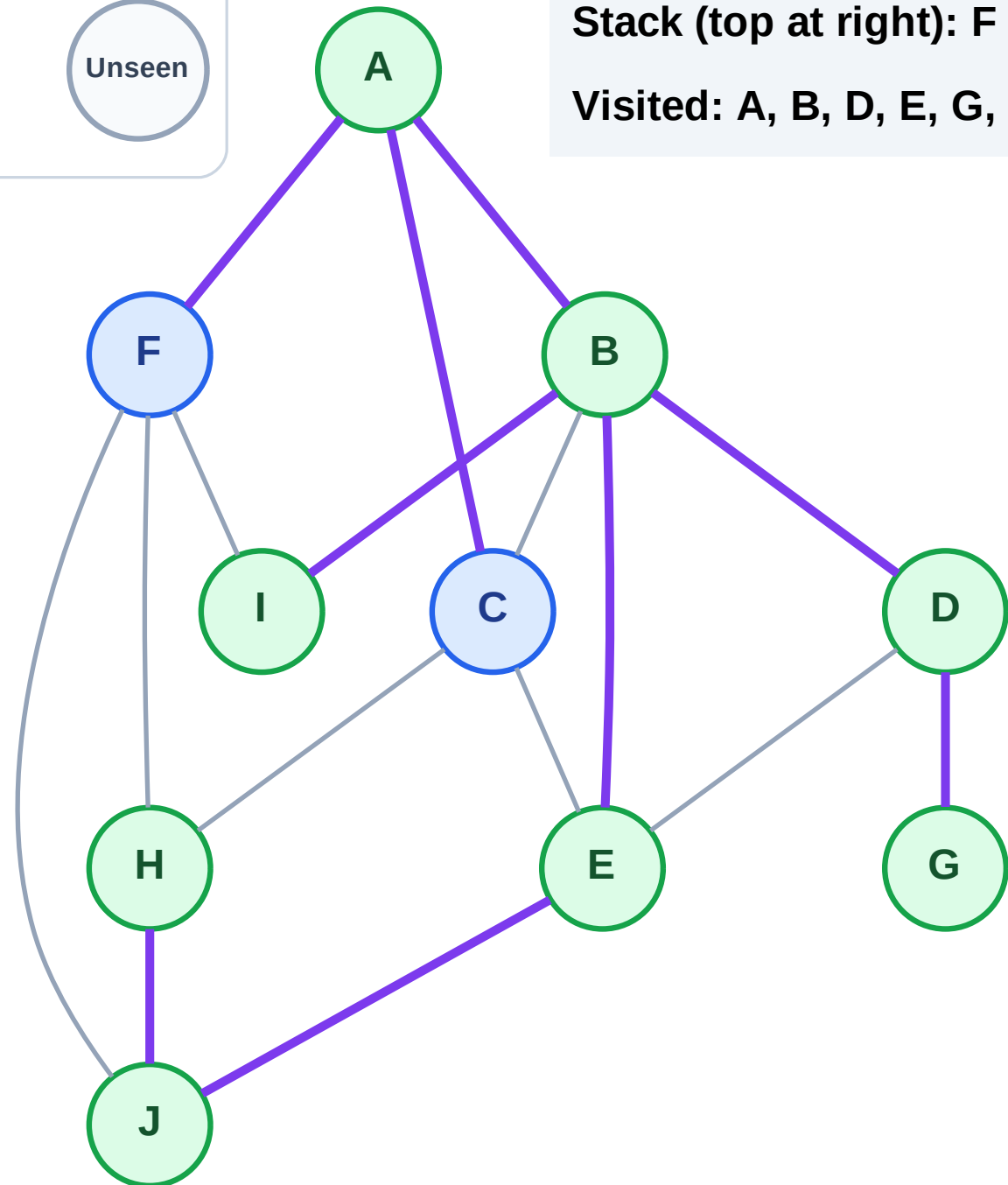
Step 17: No new neighbors from I

Legend



Stack (top at right): F | C

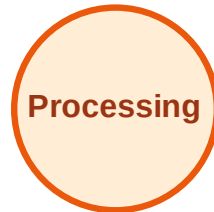
Visited: A, B, D, E, G, H, I, J



DFS Traversal

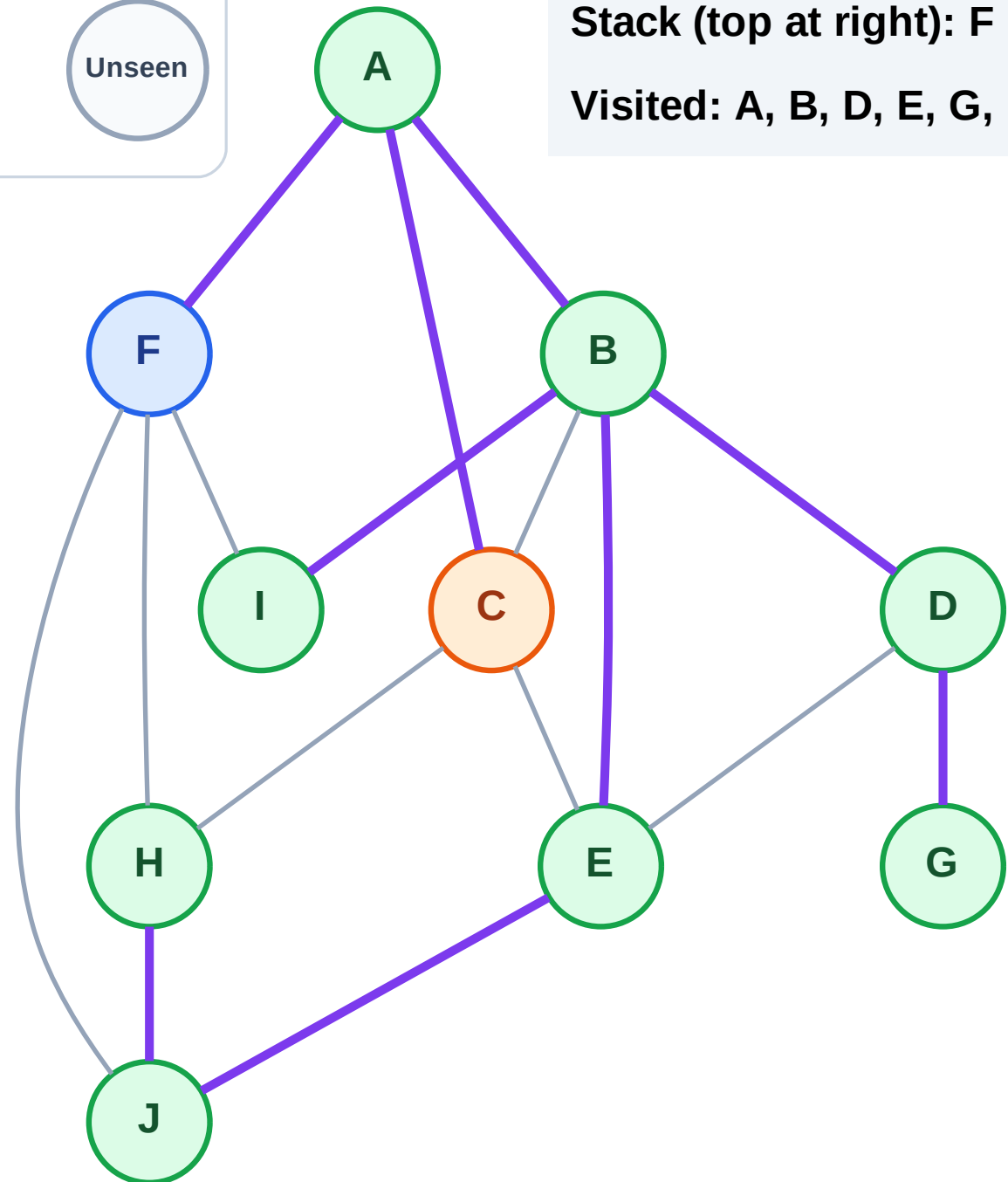
Step 18: Process node C

Legend



Stack (top at right): F

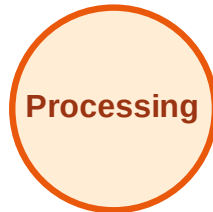
Visited: A, B, D, E, G, H, I, J



DFS Traversal

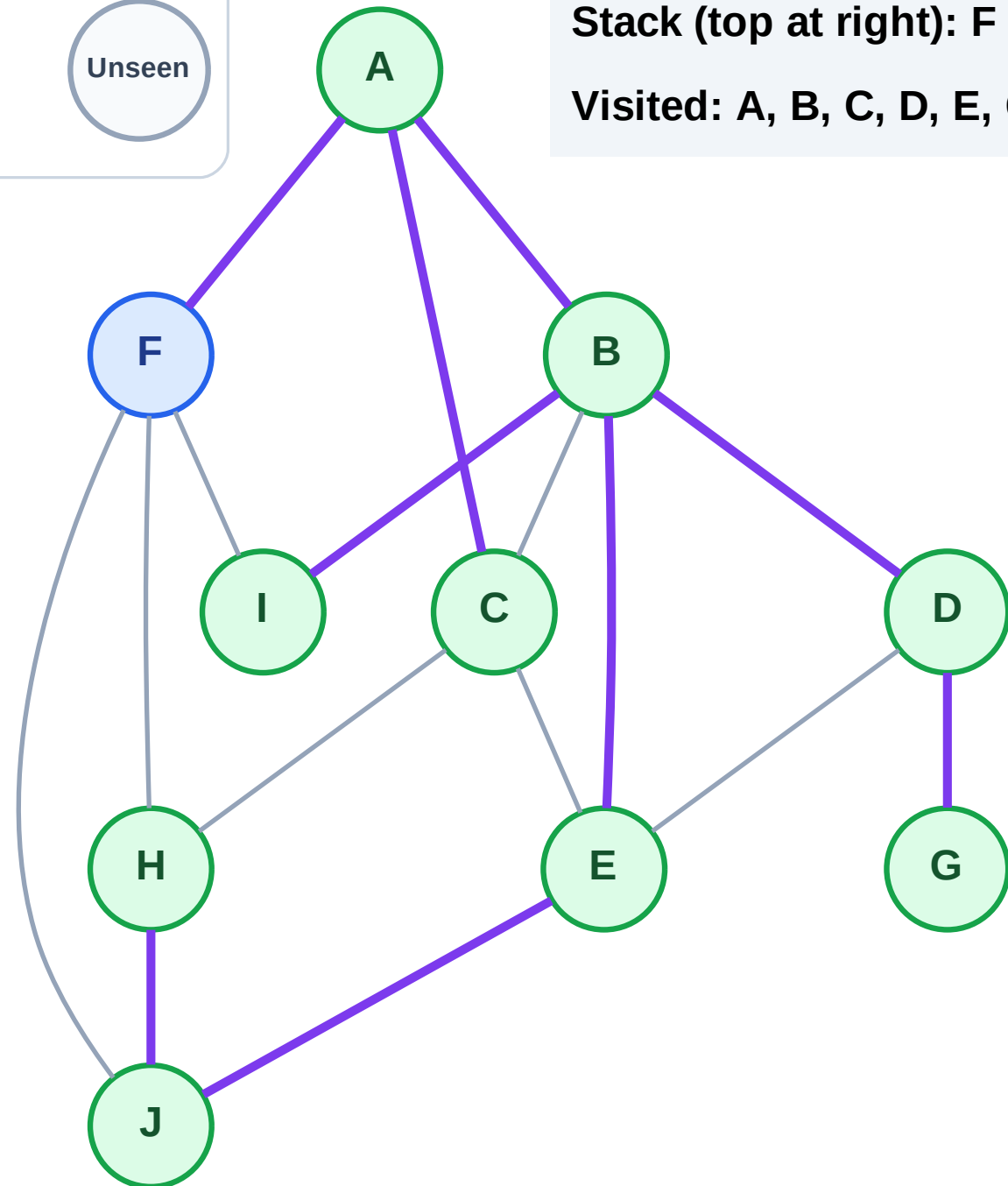
Step 19: No new neighbors from C

Legend



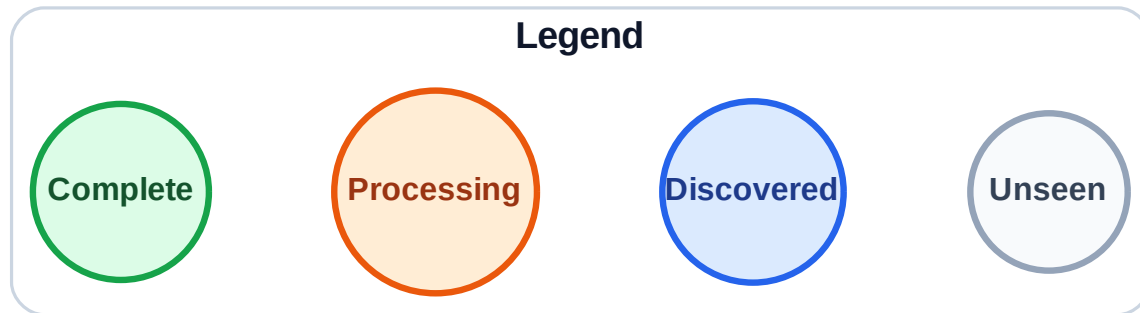
Stack (top at right): F

Visited: A, B, C, D, E, G, H, I, J



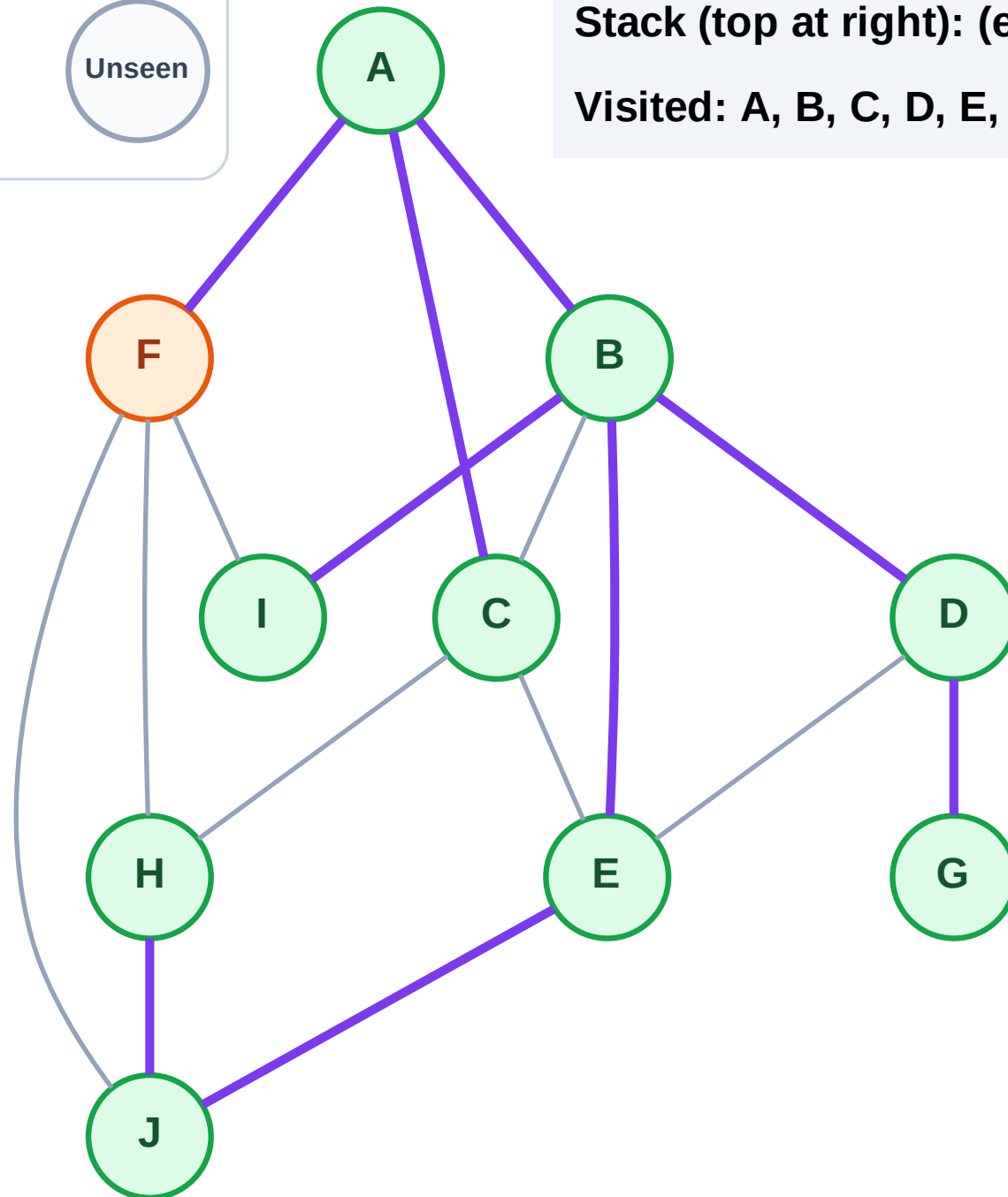
Step 20: Process node F

Step 20: Process node F



Stack (top at right): (empty)

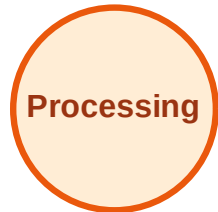
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

